

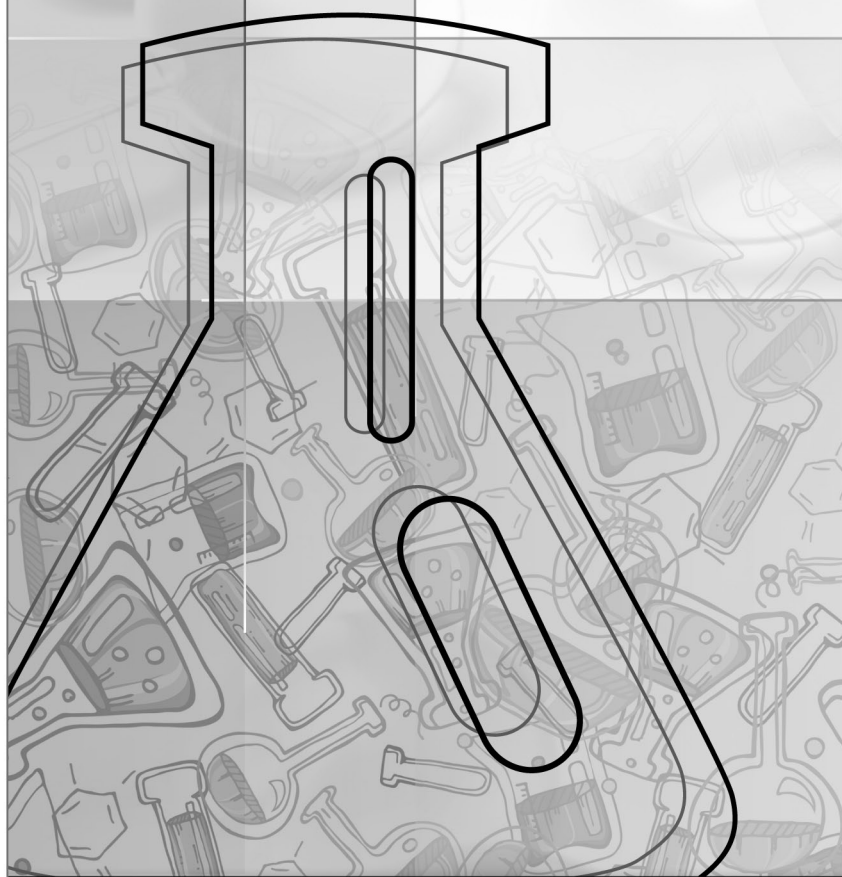
Interpreting data in **Senior Chemistry**

Grahame Dobinson



Quality ready-to-use resources

Interpreting data in **Senior Chemistry**



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Grahame Dobinson is a science teacher with many years of experience in teaching all branches of science. He has a genuine love of this subject area, and is an established and successful author who has a longstanding association with Blake Education.

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INTRODUCTION

It was once thought that two chemicals that reacted together could do so in any proportions. For example, hydrogen and oxygen could react in any proportions to form water and there would be no 'leftovers'. Experiments have subsequently shown that this is not the case. This exercise uses data from the reaction between iron and chlorine, which produces iron chloride.

Aim

To determine whether or not any mass relationships exist in the reaction between iron and chlorine.

Data

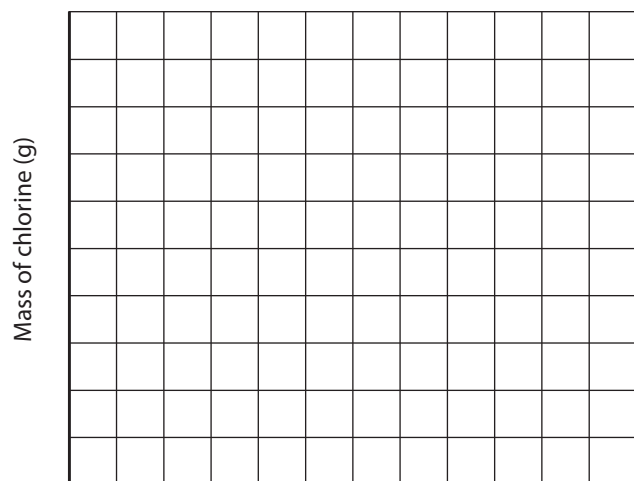
Five samples of a compound containing only iron and chlorine were prepared using different methods. Each sample was analysed and the mass of iron and chlorine in each sample was determined. The results of the analyses are given in the following table.

Sample number	Mass of iron (g)	Mass of chlorine (g)	Mass ratio of Cl to Fe
1	5.59	10.64	
2	1.68	3.20	
3	1.40	2.67	
4	1.96	3.73	
5	3.35	6.38	

Relative atomic masses: iron = 56, chlorine = 35.5

Analysing the data

- 1 Graph these results, plotting the mass of chlorine (y-axis) against the mass of iron (x-axis).



Mass of iron (g)

- 2 Calculate the ratio of the mass of chlorine to mass of iron for each sample. Enter the results in the appropriate column of the table.

Interpreting the data

- 3 Describe the shape of the graph plotted.
- 4 What does this shape indicate about the ratio of the mass of chlorine to the mass of iron in each sample?
- 5 Is the conclusion reached in question 4 consistent with the calculations entered into the table from question 2?
- 6 From the graph, predict:
- the mass of chlorine that would react with 4.00 g of iron.
 - the mass of iron that would react with 2.00 g of chlorine.
- 7 Using the data from sample 1, determine the percentage of iron, by mass, in the sample.
- 8 Would the percentage obtained for question 7 be the same for the other four samples?
- 9 The ratio of the mass of chlorine to the mass of iron is not enough to enable a formula for the iron and chlorine compound to be determined. What other information must be used in order for a formula to be deduced?

Conclusion

A choice can be made here based on the 'original ideas' of chemical combination and the ideas revealed in results such as those above. Here are the choices.

- Option 1:** Elements such as iron and chlorine can react in such a way that the masses of each are not really important. They will always react totally.
 - Option 2:** Elements such as iron and chlorine react only in definite proportions by mass.
- 10 Which option seems to be the one supported by the data?

INTRODUCTION

A few centuries ago it was starting to become apparent to experimental chemists that the reaction between chemicals was not a 'haphazard' thing, but that there seemed to be an orderly nature to these reactions, particularly in relation to the masses of reactants and products.

Aim

To analyse experimental data from a chemical reaction in which the masses of reactants and products have been recorded accurately.

Data

When hydrogen gas is passed over hot copper(II) oxide, the oxide is converted into copper metal and the hydrogen combines with the oxygen to form water. An experiment was carried out in which hydrogen gas was passed over heated copper(II) oxide in a glass tube. The water formed was carefully collected in a U tube containing a chemical that readily absorbs water. A second U tube containing this same chemical was connected to the first. The apparatus is shown in the diagram below. Mass measurements of all tubes were taken before and after the experiment.

	Tube A	Tube B	Tube C
Initial mass	31.520 g	42.630 g	41.890 g
Final mass	29.120 g	45.332 g	41.890 g

Interpreting the data

1 Write a balanced chemical equation for the reaction that occurs.

2 At the conclusion of the experiment, what new chemical is present in:

a tube A? _____

b tube B? _____

3 Determine the decrease in mass in tube A.

4 Determine the increase in mass in tube B.

5 What does the loss of mass in tube A represent?

6 What does the increase in mass in tube B represent?

7 Tube C also contained a chemical that absorbed water. What might be the reason for it not changing mass?

8 As there was no change in mass, does tube C have a role? Explain.

9 The loss of mass from tube A is not equal to the gain in mass in tube B. What is the reason for this?

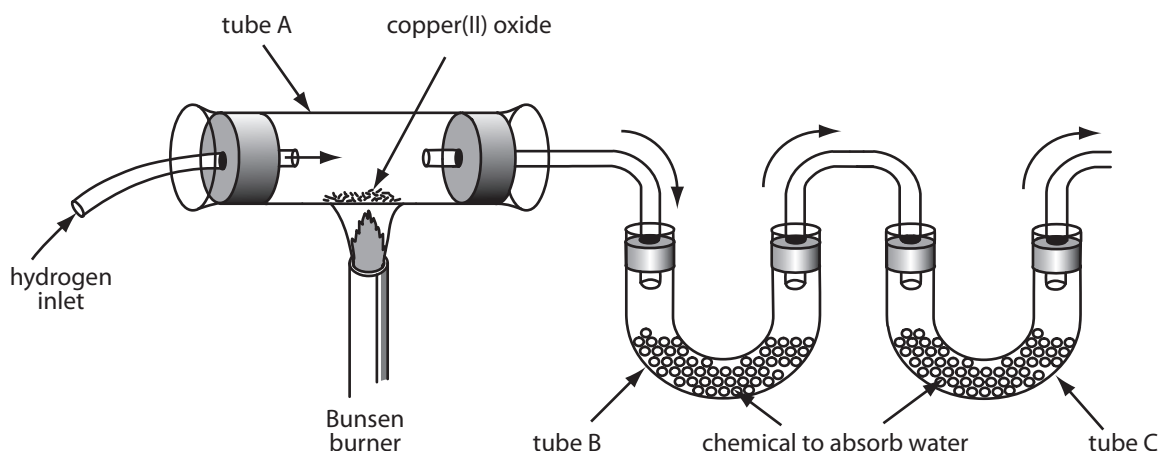
10 Given the data and considering the apparatus used for this reaction, would the reaction be classed as an exothermic or endothermic one?

11 When a compound loses oxygen as in this reaction, it is said to be 'reduced'. If a chemical gains oxygen it is said to be 'oxidised'. Both of these processes occur in this reaction. Identify the chemical that is:

a reduced. _____

b oxidised. _____

12 Would any colour change be expected to occur in tube A as the reaction proceeds? Explain.



INTRODUCTION

This exercise outlines the way in which chemists systematically find out what a chemical compound is composed of and the proportions of elements present in a particle of the compound. This gives a formula called an 'empirical formula'. The molecular formula provides more information and shows the actual number of atoms of each element present in a molecule of the compound.

Aim

To determine the empirical formula and the molecular formula of the compound benzene.

Note: The data will be presented and interpreted progressively.

Data: Step 1 Finding which elements are present in the compound

- a When benzene is burned in oxygen the reaction is:
benzene(l) + oxygen(g) → carbon dioxide(g) + water(l)
- b The proportion, by mass, of carbon in carbon dioxide is 27.3% and of hydrogen in water is 11.1%.
- c When 3.12 g of benzene is burned in oxygen, 10.56 g of carbon dioxide and 2.16 g of water are formed.

Interpreting the data

- 1 Since the benzene was burned in oxygen only, where must the carbon in the carbon dioxide and the hydrogen in the water have come from? _____
- 2 Does this mean that there is definitely no oxygen in the benzene? (Yes/No) _____
- 3 Calculate the mass of:
a carbon in the carbon dioxide. _____
b hydrogen in the water formed. _____
- 4 What is the total mass of (carbon in the carbon dioxide) + (hydrogen in the water)?

- 5 How does the total mass calculated in question 4 compare with the mass of benzene burned?

- 6 Does this mean that there is definitely no oxygen in benzene? (Yes/No) _____
- 7 What are the elements present in benzene?

Data: Step 2 Finding the proportion of each element in benzene

- d The atomic mass of hydrogen H = 1 and that of carbon C = 12.

Interpreting further

- 8 From the masses of C and H in the benzene sample, determine the percentage, by mass, of each of the elements in benzene.
a C _____
b H _____
- 9 Since C atoms are 12 times the mass of H atoms, the ratio of atoms in a particle of benzene is given by C : H = (% of C from question 8 ÷ 12) : (% of H from question 8 ÷ 1). Reduce this ratio to simple whole numbers.
C : H = _____
- 10 The empirical formula for benzene must be:

Data: Step 3 Finding the molecular formula for benzene

The molecular formula is some multiple of the empirical formula, i.e. $\times 1$, $\times 2$, $\times 3$...

- 11 From the empirical formula calculated in Step 2, what would be the mass of a molecule of benzene if it had only the number of C and H atoms in it as in the empirical formula? _____
- 12 Experiments with benzene in the vapour state show that it has a molecular mass of 78 amu.
How many multiples of the empirical formula mass is this molecular mass? _____
- 13 What, then, must be the molecular formula for benzene? _____

A summary of the findings

- 14 What elements are present in benzene?

- 15 What is the empirical formula of benzene?

- 16 Write down the molecular formula of benzene.

Additional questions

- 17 There is another formula for a compound such as benzene which needs to be known in order to understand the chemical nature of the compound. What is the name of this formula?

- 18 What does this formula show that the others do not?

INTRODUCTION

A common method of preparing solid samples of chemical salts is to evaporate the water from a solution of the particular salt, as is done with common salt, sodium chloride. However, when this is done, some water from the solution often becomes attached to the solid crystalline salt. The salt is then said to be 'hydrous' or 'hydrated' and the attached water is referred to as 'water of crystallisation'. Although the water does not change the chemical properties of the salt, it does alter other properties such as the colour and crystal shape.

Aim

To see how the presence of water in crystals is an important consideration when determining the mass of a salt sample.

Data

Although not all salts take on water of crystallisation, many do. The proportion of water in a particular salt can be shown in the formula. For example, the hydrated salt iron(II) sulfate is written as $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$. This means that for every 1 mole FeSO_4 there are 7 moles of water of crystallisation combined with it. Here four such hydrated salts are considered:

- a** barium chloride $\text{BaCl}_2 \cdot 2\text{H}_2\text{O}$
- b** sodium carbonate $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$
- c** copper(II) sulfate $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$
- d** cobalt(II) chloride $\text{CoCl}_2 \cdot 6\text{H}_2\text{O}$

The atomic masses of elements in these salts:

H = 1, C = 12, O = 16, S = 32, Cl = 35.5, Na = 23, Co = 59, Cu = 63.5, Ba = 137

Analysing the data

- 1** For each of the four salts involved, calculate the molar mass of the 'anhydrous' (= without any water) variety of the salt.

- a** barium chloride _____
- b** sodium carbonate _____
- c** copper(II) sulfate _____
- d** cobalt(II) chloride _____

- 2** Which salt has:

- a** the lowest molar mass?

- b** the largest amount of water of crystallisation?

- 3** For each salt, calculate the mass of water that is present in one mole of the hydrated form.

- a** barium chloride _____
- b** sodium carbonate _____
- c** copper(II) sulfate _____
- d** cobalt(II) chloride _____

- 4** Now determine the percentage of water in each hydrated salt.

- a** barium chloride _____
- b** sodium carbonate _____
- c** copper(II) sulfate _____
- d** cobalt(II) chloride _____

- 5** Which hydrated compound has the highest percentage of water?

- 6** The simplest way in which the water of crystallisation can be eliminated from a hydrated salt is by heating the salt. Suggest a minimum temperature for the heating.

Interpreting the data

If the presence of water of crystallisation in a salt is understood it can prevent, for example, mistakes being made when preparing solutions. Here is a case in question. A student wishes to make 500 mL of a 1.0 M solution of sodium carbonate. He buys some 'washing soda' (crystals of hydrated sodium carbonate) and calculates the mass of sodium carbonate needed for the job. His calculations suggest that he needs 53.0 g of sodium carbonate, so he weighs out this amount of his washing soda crystals, dissolves them in distilled water and makes the final volume 500 mL, thus making his desired solution. However, his procedure is flawed.

- 7** Explain, in principle, why this is. _____

- 8** Determine the actual amount of anhydrous sodium carbonate present in his 53.0 g of crystals.

- 9** Determine from the calculated value in question 8 the actual molarity of the solution he prepared.

- 10** What could he have done with the crystals to assure that he had the necessary mass of sodium carbonate for his solution?

INTRODUCTION

The noble gases constitute a group in the periodic table that is unique because of the general inactivity of the gases. There are six such gases. Of the remaining elements there are only five others that are gases at normal temperature and pressure. Table 1 presents some data about the noble gases. It has been arranged with the gases in alphabetical order, i.e. argon is first and xenon is last.

Aim

To analyse this data, look for any trends that may occur and to offer explanations where possible.

Presenting the data

To analyse this data it is better to present it with a chemical basis rather than simply in alphabetical order. The first task, then, is to rearrange the table to list the gases in order of increasing atomic number. Rearrange this data accordingly in Table 2.

Analysing the data

- Which table allows trends related to atomic number to be identified more readily?
- How do the melting points and boiling points of these elements compare, in general, with those of most other elements?
- What is the lowest single temperature to which all of these gases could be reduced and have them still all gases?
- Some additional first ionisation energies:
 $\text{Na} = 5.0 \times 10^2$, $\text{Mg} = 7.4 \times 10^2$, $\text{Al} = 5.8 \times 10^2$,
 $\text{C} = 10.9 \times 10^2$
 How do the values of the ionisation energies for the noble gases generally compare with these?

Table 1

No.	Noble gas	Atomic number	Atomic mass	Molecular mass	Melting point (°C)	Boiling point (°C)	Ionisation energy	No. of isotopes
1	Argon	18	40	40	-189	-186	15.2×10^2	3
2	Helium	2	4	4	-272 (under pressure)	-269	23.6×10^2	2
3	Krypton	36	84	84	-157	-152	13.5×10^2	5
4	Neon	10	20	20	-249	-246	20.1×10^2	3
5	Radon	86	222	222	-71	-62	10.3×10^2	0
6	Xenon	54	131	131	-112	-108	11.7×10^2	9

Notes:

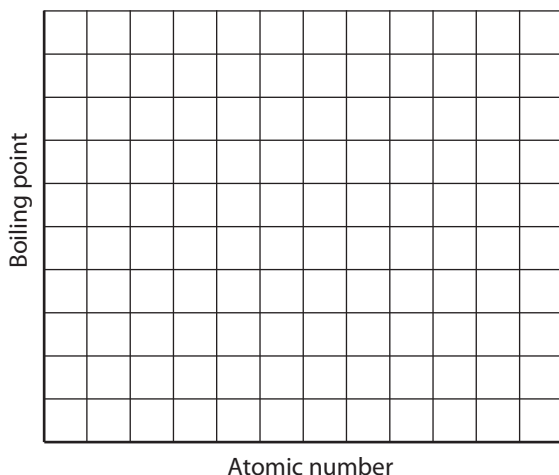
- The atomic mass, molecular mass, melting point and boiling point are rounded to whole numbers.
- The ionisation energies are the first ionisation energies and the units are kJ mole^{-1} .
- Radon is a radioactive gas.

Table 2

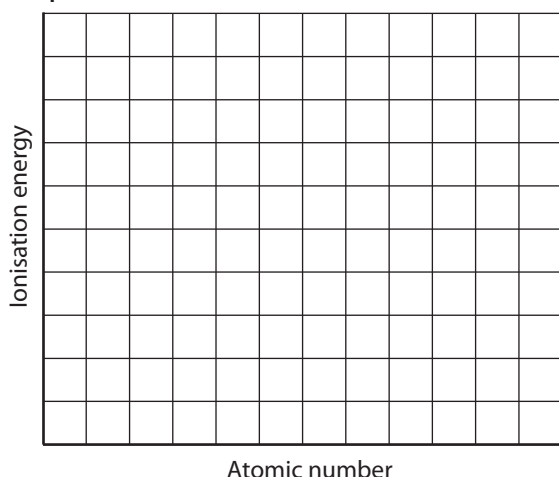
No.	Noble gas	Atomic number	Atomic mass	Molecular mass	Melting point (°C)	Boiling point (°C)	Ionisation energy	No. of isotopes
1								
2								
3								
4								
5								
6								

- 5 Construct two graphs using values from the data. Graph A is a graph of atomic no. (x-axis) vs boiling point (y-axis). Graph B is a graph of atomic no. (x-axis) vs ionisation energy (y-axis).

Graph A



Graph B



Interpreting the data

- 6 Make a simple qualitative statement to show what each graph reveals.
- a Graph A _____
- b Graph B _____
- 7 Explain what the difference between a high first ionisation energy and a low first ionisation energy means about atoms of various elements.
- _____
- _____
- 8 The data shows that radon has no isotopes, unlike the other noble gases. Is this solely because radon is radioactive? (Yes/No) _____
- Justify your answer. _____
- _____

- 9 Try to name the other five elements that are also gases at room temperature. _____
- _____
- 10 The molecular masses of the five other elements that are gases at room temperature are equal to twice their atomic masses, but Tables 1 and 2 (page 8) show that the molecular mass of each noble gas is the same as its atomic mass. How can this be explained?
- _____
- _____
- 11 Which noble gas is the one from which it is hardest to extract a first electron from one of its atoms?
- _____
- 12 Which noble gas would have the weakest bonds holding it together in the solid state?
- _____
- 13 No stable compound of helium has been found, but many compounds of xenon have been synthesised. Suggest a reason for this.
- _____
- _____
- 14 Without constructing a detailed graph, draw an approximate line graph to predict the shape of a graph of first ionisation energy (x-axis) versus BP (y-axis) for the noble gases.
-
- Boiling point
- First ionisation energy
- 15 In terms of modern atomic structure, explain why the noble gases are generally so inactive chemically.
- _____
- _____
- _____
- 16 What was the older name for these gases, which was used before any compound of any of the noble gases was found?
- _____

INTRODUCTION

Hydrocarbons are compounds of only two elements, hydrogen and carbon. There are three such compounds containing only two carbon atoms per molecule. These are called ethane, ethene and ethyne. Some of the properties of these compounds are listed in the table below.

Aim

To identify and explain any trends that occur in the properties of the hydrocarbon compounds that have two carbon atoms in each molecule.

Data

Properties	Ethane	Ethene	Ethyne
Empirical formula	CH ₃	CH ₂	CH
Molecular formula	C ₂ H ₆	C ₂ H ₄	C ₂ H ₂
Molecular mass	30	28	26
Boiling point (K)	185	171	198
Carbon to hydrogen bond lengths (nm)	0.110	0.110	0.106
Carbon to carbon bond lengths (nm)	0.154	0.135	0.121
Carbon to carbon bond energies (kJ mol ⁻¹)	347	612	832

Analysing the data

- What is the trend in the number of H atoms per molecule across the table from left to right?

- What is the trend in the molecular mass across the table from left to right?

- Is there any trend in the C to H bond lengths between the three hydrocarbons in the table?

- Is there any significant trend in the C to C bond lengths as the number of H atoms per molecule decreases?

Interpreting the data

- What appears to be the reason for the drop in molecular mass across the table from ethane to ethene to ethyne?

- What would be the most likely reason for the trend in the lengths of the C to C bonds in the three hydrocarbons?

- Explain the fact that there is almost no difference in the C to H bond length in any of the three compounds.

- What relationship appears to exist between the C to C bond lengths and the C to C bond energies in the three compounds?

- Only one of these hydrocarbon molecules can properly be described as a *linear molecule*. Which one is it?

- Draw electron-dot diagrams to show the bonding that occurs between atoms in each of the three molecules.
 - Ethane
 - Ethene
 - Ethyne

INTRODUCTION

Many of the more simple organic compounds have relatively low boiling points and so are commonly gases or liquids at room temperature. The table below lists the boiling points for some alkanes, alkanols and alkanolic acids.

Aim

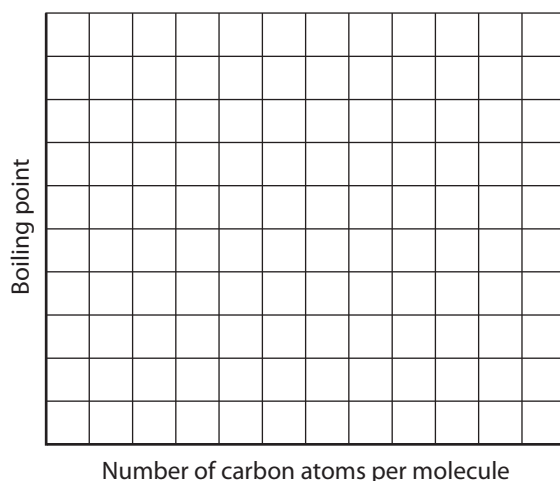
To determine if any trends occur in the boiling points of some organic compounds and to try to explain such trends.

Data

Note: None of the compounds listed in the table is a solid at room temperature.

Analysing the data

- I Draw three line graphs on one set of axes to plot these data. Put 'Number of carbon atoms per molecule' on the x-axis and 'Boiling point' on the y-axis.



- 2 What name is given to a number of organic compounds whose various formulae can be represented by a single 'general formula'?

- 3 What is the general formula for each of these organic chemical groups?

a Alkanes: _____

b Alkanols: _____

c Alkanolic acids: _____

- 4 Within the alkane group, what association can be seen between the number of C atoms in the molecule and the BP of the compound?

- 5 Does this trend apply to:

a the alkanols? _____

b the alkanolic acids? _____

Interpreting the data

- 6 Complete the table by filling in the 'State at room temperature (25°C)' column for each compound.
- 7 Given that the data shows that propane is normally a gas at room temperature, how is it possible that propane is usually transported in containers in the liquid state? _____
- 8 Alkanes such as propane and butane are, generally, quite inert chemically. They do not react with common reagents such as acids and alkalis. However, there are quite strict safety procedures that apply to their transportation and use. Why is this? _____
- 9 The graphs should show that each alkanol has a BP generally well above that for the alkane with the same number of carbon atoms. In turn, the BP of the alkanolic acid is above that of the corresponding alkanol. Explain why this is thought to occur. _____

Alkane	BP (°C)	State at room temperature (25°C)	Alkanol (Primary)	BP (°C)	State at room temperature (25°C)	Alkanolic acid	BP (°C)	State at room temperature (25°C)
Methane	-160		Methanol	65		Methanoic	101	
Ethane	-88		Ethanol	79		Ethanoic	118	
Propane	-41		Propanol	97		Propanoic	141	
Butane	1		Butanol	117		Butanoic	164	
Pentane	38		Pentanol	138		Pentanoic	186	
Hexane	69		Hexanol	157		Hexanoic	205	

INTRODUCTION

The Earth's atmosphere contains a variety of gases and so is a rich resource if used wisely. There are also particles of solid substances such as dust which are not considered to be 'resources'. A particular gas can be separated from the mixture if its properties are known.

Aim

To investigate some properties of the more abundant gases in the Earth's atmosphere and see how this knowledge can assist in the separation of some of the more sought after ones.

Note: Although water vapour is present in the atmosphere it is not considered in this exercise.

Data

Some properties of the five most abundant atmospheric gases are presented in the table below.

Analysing the data

- Fill in the last column by calculating the relative vapour density of each gas.
- What is meant by the term 'sublimes'?

- What is meant by a substance 'hydrolysing' in water?

Interpreting the data

- If the temperature of a sample of air in a container were to be lowered gradually, which gas would be:
 - the first to change state? _____
 - the last to change state? _____

- If the temperature of the air sample were to be lowered to -187°C , how many gases would change state? _____

Name these gases. _____

- Solid carbon dioxide is often used as a type of 'ice' for cooling. It is called 'dry ice'. With reference to the table, suggest a reason for this.

- If a sample of a gas mixture containing some of the gases in the table was suspected of containing carbon dioxide, what test could be used to detect its presence?

- Rank the gases in the table from the least dense to the most dense.

- Although most of the gases in the atmosphere do not burn or support combustion, carbon dioxide is generally chosen for use in fire extinguishers. From the information in the table suggest a reason for this.

No.	Gas	Molecular mass	MP ($^{\circ}\text{C}$)	BP ($^{\circ}\text{C}$)	Effect on combustion	Reaction with $\text{Ca}(\text{OH})_2$	Solubility in water (units)	Relative vapour density
1	Oxygen	32	-219	-183	Assists combustion	No reaction	0.04	
2	Nitrogen	28	-210	-196	Does not assist	No reaction	0.02	
3	Carbon dioxide	44	*	-78	Does not assist	Forms a white solid	1.69 and hydrolyses	
4	Argon	40	-189	-186	Does not assist	No reaction	0.06	
5	Neon	20	-249	-246	Does not assist	No reaction	0.01	

Notes:

* Carbon dioxide sublimes rather than melting and then evaporating.

The relative vapour density of a gas is a measure of the density of the gas compared with that of hydrogen, which is given a value of 1. It is calculated by halving the molecular mass of the gas.

INTRODUCTION

Modern jet aircraft, particularly those transporting large numbers of people or with large cargoes, use quite substantial amounts of atmospheric oxygen to burn the necessary fuel for their flights. As a result, they produce substantial amounts of carbon dioxide and heat energy.

Aim

To investigate the statistics of a single long-distance jet flight to determine the total amounts of (a) fuel burned, (b) oxygen consumed and (c) carbon dioxide produced.

Data

Note: Much of the data has been simplified.

This is data from which information can be selected for answering the later questions.

- a** The air around us is approximately 20% oxygen.
- b** Oxygen is put into the air by the process of photosynthesis in green plants. This is represented by the following equation:

$$6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$$
- c** When a hydrocarbon fuel burns completely it produces water and carbon dioxide.
- d** The composition of aviation fuel varies considerably, but it is always a mixture of hydrocarbons, mostly alkanes. These vary in carbon chain length from C_5 to C_{15} . The hydrocarbon decane $\text{C}_{10}\text{H}_{22}$ is an 'average' of these.
- e** The density of decane is 0.73 g mL^{-1} at 20°C and its MP is -27°C ; its BP is 174°C .
- f** A laden Boeing 767 uses approximately 4.7 litres of fuel every second while in flight.
- g** Assume that a flight in a 767 from Sydney to Los Angeles takes 13.5 hours.

Analysing the data

- 1** Write a balanced equation for the complete combustion of decane.

- 2** Determine the mass of one litre of decane.

- 3** What is the molar mass of decane?

- 4** How many moles of decane are present in a litre of the hydrocarbon?

Interpreting the data

Note: For this section decane can be assumed to be an 'average' formula for aviation fuel.

- 5** Calculate the number of litres of decane burned on a Sydney to Los Angeles flight.

- 6** What mass, in grams, does the volume in question 5 represent?

- 7** Using the balanced equation from question 1, calculate the mass of oxygen required to burn the mass of decane in question 6.

- 8** Calculate the mass of carbon dioxide produced from the above combustion.

- 9** Considering that 44.0 g of CO_2 would occupy a volume of 22.4 L at 0°C and 1.0 atm pressure, what would be the volume of CO_2 in question 8 under the same conditions?

- 10** Considering the altitude of international flights, what is one reason for having to correct the value calculated in question 9 to make it a more realistic figure for the volume of CO_2 produced during such a flight?

INTRODUCTION

The sand that occurs on Australia's beaches varies slightly in its chemical composition from beach to beach and depends on the local environment, both ancient and modern, in regard to its makeup. For example, sand from the beaches in NSW may contain coal in minute amounts while samples taken from beaches in WA or SA may have none of this. Sand is a valuable source of a number of minerals.

Aim

To interpret the analysis of a particular sample of an Australian beach sand.

Data

Atomic masses of some elements:

C = 12, O = 16, Si = 28, Ca = 40, Fe = 56, titanium Ti = 48, zirconium Zr = 91

Interpreting the data

- Five of the compounds listed in Table 1 have very specific chemical formulae while one does not. Explain why this is the case.

- Write an appropriate chemical name for each of the first five substances in Table 1.
- Fill in the solubility column in Table 2 using the words 'soluble' and 'insoluble' and considering where sand is found.
- Would filtration be a suitable method for separating any of these components? _____
Explain. _____

- Quartz and calcite have some very similar properties. What simple test would distinguish between them?

- From the formulae in Table 1 state the valency of:
 - silicon. _____
 - titanium. _____
 - iron. _____
 - zirconium. _____
- Which two components would be hardest to separate according to the physical properties in the table?

- If a sample of the sand was placed in an unreactive liquid of density 2.8 g mL^{-1} , would any separation of the sand's components take place? _____
Explain. _____
- Most of the components in beach sand are the 'leftovers' of the weathering of rocks. What could be another source of the mineral calcite in beach sand?

- Calculate the percentage, by mass, of titanium in rutile.

- If a sample of beach sand contained 0.5% rutile, what mass of titanium could be extracted from 1.0 kg of the sample?

Table 1 Sample analysis results

Component: mineral name	Quartz	Calcite	Rutile	Ilmenite	Zircon	Coal
Chemical formula	SiO_2	CaCO_3	TiO_2	$\text{FeO} \cdot \text{TiO}_2$	ZrSiO_4	$\approx \text{C}$ (mainly)
Chemical name						–
Major/minor component	Major	Minor	Minor	Minor	Minor	Minor

Table 2 Some properties of the components

Component	Solubility	Density (g mL^{-1})	Colour	Reaction with HCl
Quartz		2.65	Colourless if pure	None
Calcite		2.71	Colourless/white	Reacts vigorously
Rutile		4.2	Red/brown	None
Ilmenite		4.5	Black	None
Zircon		4.7	Colourless	None
Coal		≈ 1.5	Black	None

INTRODUCTION

Coal has long been an important part of the Australian economy. Australia is the world's leading coal exporter. Coal varies considerably in its chemical composition. Specific names are given to the various types on the basis of their approximate chemical composition, specifically of their carbon contents. Whatever the composition of a particular coal, its main constituent is carbon. This exercise deals with the three types of coal that are important to Australia.

Aim

To look at some of the chemistry of samples of three types of coal and at the effects of their combustion.

Data

Note: The values listed in Table 1 are approximate.

Interpreting the data

- 1 Write balanced equations for the combustion of the following. Assume that in the combustion of carbon, nitrogen and sulfur only the dioxides are formed.
 - a carbon _____
 - b hydrogen _____
 - c nitrogen _____
 - d sulfur _____
- 2 In 100 g of anthracite coal, what is the mass of each component?
 - a carbon _____
 - b hydrogen _____
 - c nitrogen _____
 - d sulfur _____
- 3 Suppose that 100 g of anthracite coal is completely burned. Calculate the mass produced of:
 - a carbon dioxide. _____
 - b sulfur dioxide. _____
 - c nitrogen dioxide. _____

- 4 Determine the mass of oxygen needed for the total combustion of the elements in question 3.

- 5 If a coal-exporting port had an annual output of, say, only one-quarter of that of Newcastle and all of the coal was anthracite, calculate the total mass of carbon dioxide produced if it were all burned.

- 6 If some of the coal considered in question 5 were of the bituminous type, how would this affect the proportionate amounts of the four oxides produced by combustion? For example, would the amounts increase or decrease?
 - a CO_2 _____
 - b H_2O _____
 - c NO_2 _____
 - d SO_2 _____
- 7 Each of the gases carbon dioxide and nitrogen dioxide dissolves in water. What is the nature of the solutions which they form, i.e. acidic or basic?

- 8 When burned, which types of coal would produce:
 - a the least amount of nitrogen dioxide?

 - b the greatest amount of nitrogen dioxide?

*Gas volume calculations can be included if needed.

Table 1 Composition of coal

No.	Type of coal	Carbon (%)	Hydrogen (%)	Nitrogen (%)	Sulfur (%)	Oxygen (%)
1	Lignite (brown coal)	71.0	5.2	1.3	2.4	20.1
2	Bituminous	84.2	5.0	1.2	1.0	8.6
3	Anthracite	93.1	2.4	1.0	0.8	2.7

Atomic masses: H = 1, C = 12, N = 14, O = 16, S = 32

Export volume of coal from the port of Newcastle, NSW, per year $\approx 80 \times 10^6$ tonnes.

INTRODUCTION

The heat of combustion (ΔH_c) of a substance is defined as the amount of heat energy given out when one mole of substance is completely burned. It is usually expressed in kJ mol^{-1} or kcal mol^{-1} .

Aim

To compare some data on the combustion of a few compounds which are commonly used as fuels and to see if any relationships exist between them.

Data

Table 1 shows the heats of combustion of some compounds.

Organising the data

- Complete Table 1.
- In Table 2 write a balanced molecular equation for each of the five fuels. Then carry out any necessary calculations to complete the four remaining columns for the five fuels.

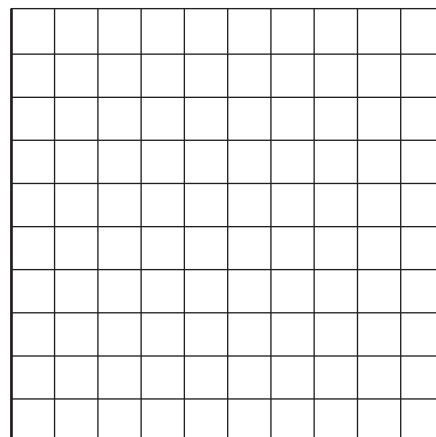
Analysing the data

To do this, draw graphs 1–4.

a

ΔH_c
 kJ mol^{-1}

Graph 1



b

ΔH_c
 kJ g^{-1}

Molar mass of fuel

Graph 2

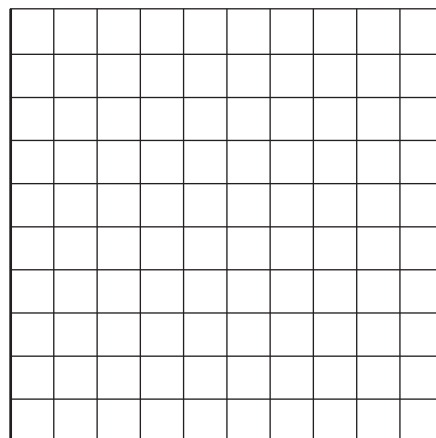


Table 1

No.	Compound (fuel)	Molecular formula	Molar mass (g)	ΔH_c (kJ mol^{-1})	ΔH_c (kJ g^{-1})
1	Methane	CH_4		890	
2	Ethane	C_2H_6		1560	
3	Propane	C_3H_8		2220	
4	Butane	C_4H_{10}		2877	
5	Pentane	C_5H_{12}		3509	

Table 2

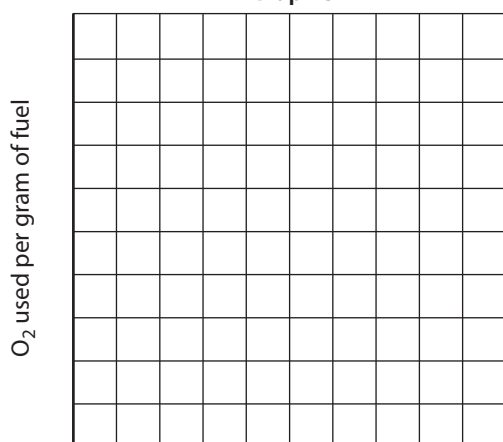
Note: (a) The numbers in Table 2 refer to the compounds with the same number in Table 1.

(b) The states of reactants and products may be left out of the equations.

No.	Equation for combustion of 1 mole of fuel	Mass O_2 used (g)	Mass CO_2 produced (g)	Mass O_2 used per gram of fuel (g)	Mass CO_2 produced per gram of fuel (g)
1					
2					
3					
4					
5					

c

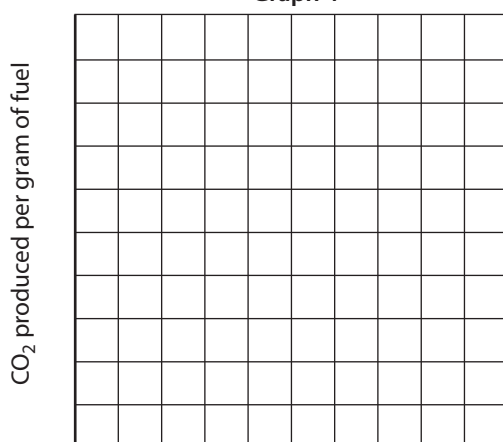
Graph 3



Molar mass of fuel

d

Graph 4



Molar mass of fuel

- 3 To what homologous series do the five hydrocarbons listed in Table 1 belong?

- 4 Examination of the successive molar masses shows a constant increment of 14.0 g. Explain why this occurs.

Interpreting the data

- 5 The following questions relate to the graphs that have been drawn. In each case complete the sentence provided by observing the relevant graph.

a Graph 1 shows that as the molar mass of the fuel increases the heat of combustion per mole

b Graph 2 shows that as the molar mass of the fuel increases the heat of combustion per gram

- c Graph 3 shows that the mass of O_2 used in burning 1 g of fuel increases as the

- d Graph 4 shows that the mass of CO_2 produced when 1 g of fuel is burned decreases as the

- 6 The source of energy in any exothermic reaction is the energy released by breaking chemical bonds and forming new ones that store less energy. What does this mean about the energy stored in CO_2 and H_2O compared with that stored in a hydrocarbon fuel and O_2 ?

- 7 Which elements are bonded together in almost all hydrocarbons?

a _____ + _____

b _____ + _____

- 8 The energies stored in chemical bonds are called 'bond energies'. The bond energy of the C–C single bond is 348 kJ mol^{-1} and the bond energy of the C–H bond is 412 kJ mol^{-1} .

If a C–C bond and a C–H bond were broken, which bond would release more energy?

- 9 How would this help to explain the fact that methane has the highest heat of combustion per gram of the five hydrocarbons?

- 10 a Which fuel produces the most energy per gram?

b Which fuel produces the least amount of carbon dioxide per gram?

INTRODUCTION

The term 'enthalpy' relates to the amount of heat energy stored in a particular substance or in a group (system) of substances. In a chemical reaction, as new substances are formed the enthalpy of each substance changes and so does the enthalpy of the system. Sometimes energy is put into the system, sometimes it leaves the system.

Aim

To use some known enthalpy changes to assist in the prediction of enthalpy changes for some other reactions, in the context of the Law of Conservation of Energy.

Data

a The direct way of measuring the energy change in a chemical reaction (R) is to carry out the reaction and take appropriate measurements. Another way, avoiding the practical issue, is to use other reactions that, when added together, give the desired reaction. In this case, the energy data from these reactions can also be added together to give the energy change for the desired reactions. That is, if $R_1 + R_2 \rightarrow R_3$, then $\Delta H_1 + \Delta H_2 \rightarrow \Delta H_3$.

b Some chemical reactions and their enthalpy changes are listed in the table.

Note: These values are 'rounded' to whole numbers.

No.	Chemical reaction	ΔH (kJ)
1	$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l)$	-286
2	$C(s) + \frac{1}{2}O_2(g) \rightarrow CO(g)$	-111
3	$CO(g) + \frac{1}{2}O_2(g) \rightarrow CO_2(g)$	-284
4	$C(s) + O_2(g) \rightarrow CO_2(g)$	-394
5	$C(s) + 2H_2(g) \rightarrow CH_4(g)$	-75
6	$H_2O(l) + \frac{1}{2}O_2(g) \rightarrow H_2O_2(l)$	+98
7	$H_2(g) + O_2(g) \rightarrow H_2O_2(l)$	

c In using the data it should be remembered that if a chemical reaction is doubled, then the value of ΔH is also doubled. If a chemical reaction is reversed, then the sign of ΔH is also reversed.

Some basic questions

- What does the ' Δ ' symbol mean in ΔH ?

- What does ΔH represent about the reaction as written?

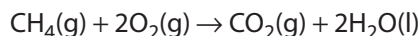
- Explain the significance of the '+' and '-' signs for ΔH .

Interpreting the data

- A simple example is to select equations from the table which, when manipulated and combined, will give equation 7. These equations are:

- The enthalpy change for equation 7 would be:

- Equations 5, 4 and 1 can be manipulated to give the equation:



Show how this can be done and hence calculate the value of ΔH for this equation.

- What does the sign and size of ΔH indicate about the reaction in question 6?

- This method of manipulating equations to find a ΔH value for a reaction has some advantages. Suggest one.

- Fuels are burned to provide energy. What sign would be expected for ΔH values in such reactions?

- Name three substances in the table which are used as fuels.

14 Chemical equilibrium

INTRODUCTION

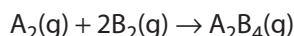
Any chemical reaction in a 'closed system' does not go fully to completion. Instead, it reaches a point where the rate of the forward reaction is equal to the rate of the reverse reaction. Nothing appears to be happening. At this stage 'chemical equilibrium' is said to be achieved.

Aim

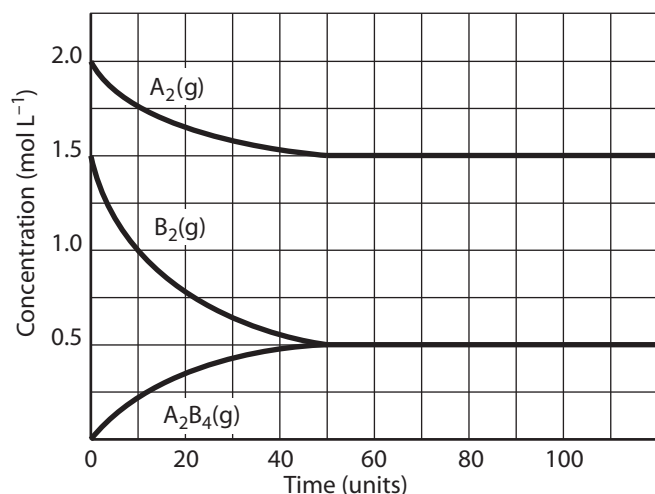
To interpret some data relating to a gaseous reaction that achieves equilibrium.

Data

Two diatomic gases, A_2 and B_2 , are known to react according to the equation:



The reaction is reversible and exothermic. Samples of the two elemental gases are introduced into a 2.0 L container and allowed to react and achieve equilibrium. The graph below plots the concentration of the gases (mol L^{-1}) vs the time (units) as the reaction proceeds.



Analysing the data

From the graph, determine the necessary information to complete the table below.

Interpreting the data

- I According to the graph, after how many 'units' of time was equilibrium achieved?

Amount of gases	Moles of $A_2(g)$	Moles of $B_2(g)$	Moles of $A_2B_4(g)$	Conc. of $A_2(g)$	Conc. of $B_2(g)$	Conc. of $A_2B_4(g)$
Initially						
At equilibrium						

- 2 Write an equation for the equilibrium constant for the reaction as written.

- 3 Using the equation from question 2, calculate K , the equilibrium constant for the reaction.

- 4 The calculated value of K indicates a reaction that goes:

- A** almost completely to the right.
B almost completely to the left.
C neither to the right or left to a great degree.

(Select A, B or C.) _____

- 5 If the original reaction was carried out at a temperature of 400°C , what effect would there be on the production of $A_2B_4(g)$ if the reaction was repeated at a temperature of:

a 300°C ? _____

b 500°C ? _____

- 6 How would the changes in temperature in question 5 affect the value of K in:

a? _____

b? _____

- 7 Considering temperature alone, if the gas A_2B_4 was a desired product for use in the further production of other chemicals, which of the three reaction temperature options in question 5 would produce the best yield of the gas?

- 8 If the pressure on the gases at equilibrium were to be increased, what effect would this have on the equilibrium and on the resulting K value?

INTRODUCTION

The term 'dissociation' is one that is usually used in chemistry to describe the 'breaking apart' of, mainly, ionic compounds, particularly when they are placed in water. Water itself does this to a small degree even though it is not classified as an ionic compound. The equation representing this is:



Aim

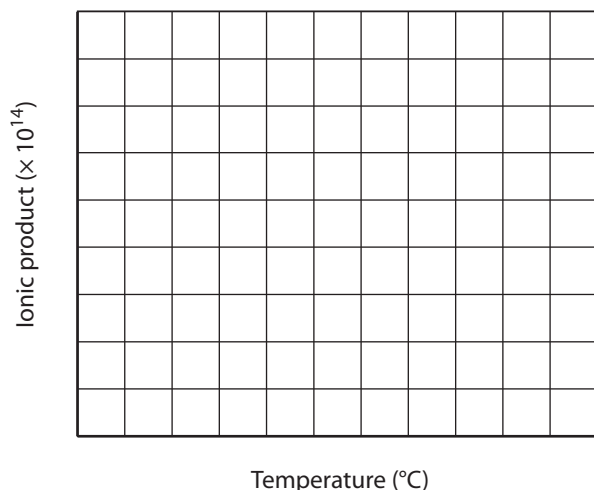
To analyse some data relating to the extent of the dissociation of water at different temperatures.

Data

The table below shows the ionic product for water at various temperatures. The ionic product K_w is defined as $K_w = [\text{H}^+] \times [\text{OH}^-]$ and typically has values of about 10^{-14} .

Analysing the data

- 1 Using the data in the table below, construct a graph of temperature vs ionic product, with temperature on the x-axis.



- 2 What does $[\text{H}^+]$ mean?

- 3 Calculate the $[\text{H}^+]$ for each of the temperatures and complete the table.

- 4 Suggest why the data does not go outside the range 0°C to 100°C .

Interpreting the data

- 5 Make a qualitative statement about the change in K_w as the temperature increases.
- 6 Does the graph show that there is a definite, direct relationship between the two variables? (Yes/No)
- 7 From the graph, estimate the value of K_w at 35°C .
- 8 At what temperature (approx.) would the value of K_w be 10.0×10^{-14} ?
- 9 Is it true to say that as temperature increases the $[\text{H}^+]$ also increases?
- 10 Does the change in $[\text{H}^+]$ with temperature mean that the neutrality of water will also change with temperature? Explain.
- 11 What does the size of the values of K_w over the range of temperatures indicate about the general extent to which water dissociates or is ionised?
- 12 The electrical conductivity of a solution or liquid depends upon the availability of ions to conduct electrical charge. Do the values of K_w indicate that water would be a good conductor or a poor conductor of electrical charge?
- 13 What is the implication from the data in the table about the ability of water to conduct electrical charge as the temperature is increased?
- 14 The table shows that the extent of dissociation of water molecules increases with increase in temperature. Explain why this occurs, in relation to the energy of the water molecules.

Temperature	0°C	10°C	20°C	25°C	30°C	40°C	50°C	100°C
Ionic product ($\times 10^{14}$ for simplicity)	0.11	0.29	0.68	1.01	1.47	2.92	5.60	51.3
$[\text{H}^+]$								

INTRODUCTION

The concept of the 'ionic product' for water is a way of measuring the extent to which water forms ions. This extent is very low. The extent of the dissociation of other substances can be measured in a similar manner, particularly for ionic binary compounds that, like water, form only two ions when they dissociate. This dissociation occurs when the binary compound is put into water which acts as a solvent. The product of the concentration of the two ions formed is given a different name. It is called the 'solubility product' as its value is an indication of how soluble the compound is. The value is achieved when the solution concerned is saturated.

Aim

To interpret some solubility product data for a group of ionic binary compounds.

Data

The table below shows the values of the solubility products for seven binary compounds. The values were determined at 25°C. The ionic product for water at 25°C is $K_w = 1.0 \times 10^{-14}$. It represents the extent to which water is ionised according to the equation $\text{H}_2\text{O} \rightarrow \text{H}^+ + \text{OH}^-$ and is defined as $K_w = [\text{H}^+] \times [\text{OH}^-]$. The value of K_w increases with temperature. The solubility product is given the symbol K_{sp} .

Analysing the data

1 Explain the meaning of the word 'solvent'.

2 What is a 'saturated' solution?

3 By comparison with the example for water, write equivalent expressions for the solubility product of each of the seven compounds listed in the spaces provided in the table.

4 Each of the compounds listed consists of a metal and a non-metal. What general name is given to such chemical compounds?

5 Rank these compounds in order from the most soluble to the least soluble, using the numbers in the table.

6 Even though the metals in these compounds do not all have the same valency, they always seem to form cations and anions in the same ratio (i.e. 1:1 in these examples). Suggest a reason for this.

7 Would this ratio be the same for a binary compound such as arsenic(III) sulfide? _____

Explain. _____

8 The compound copper(I) bromide is classed as an insoluble compound. What would this mean about the other compounds in the table?

9 Which compound in the table is:

a the least soluble? _____

b the most soluble? _____

10 The presence of ions in a solution generally adds to the solution's ability to conduct electricity. On the basis of this information, which compound would be:

a the best conductor of electricity?

b the worst conductor of electricity?

No.	Compound	Formula	Solubility product	Expression for solubility product
1	Cadmium sulfide	CdS	8×10^{-27}	
2	Iron(II) sulfide	FeS	5×10^{-18}	
3	Cobalt(II) sulfide	CoS	4×10^{-21}	
4	Copper(I) bromide	CuBr	5×10^{-9}	
5	Copper(I) iodide	CuI	1×10^{-12}	
6	Copper(II) sulfide	CuS	4×10^{-36}	
7	Silver chloride	AgCl	2×10^{-10}	

INTRODUCTION

Copper is a much sought after metal and can be extracted from a number of ores that occur naturally in the Earth's crust. Each ore is a definite chemical compound. The mineralogical name is different from the chemical name. There are quite a few stages in the overall process of converting the ore to very pure copper metal. Some of the basics of these stages are dealt with in this exercise.

Aim

To investigate the principal processes involved in the extraction of copper metal from its ores.

Data: Set 1 The physical processing

The first stage of extraction involves the breaking up of the ore and gangue (unwanted material associated with the ore) and the subsequent crushing of the ore to, usually, a fine powder.

Interpreting the data

- 1 The crushing stage requires a large amount of energy. There must be some advantage in having the particles very small. What advantage would this probably be in:

a the physical processing of the ore?

b the chemical processing of the ore?

Data: Set 2 Some of the ores of copper

The following table lists some of these copper ores and the accepted formula for each ore. Ores for which the formula includes $\text{Cu}(\text{OH})_2$ are usually referred to as 'basic' copper compounds.

Atomic masses: H = 1, C = 12, O = 16, S = 32, Si = 28, Fe = 56, Cu = 63.5

No.	Name of ore	Chemical formula	% of Cu in ore
1	Chalcopyrite	CuFeS_2	34.6
2	Chalcocite	Cu_2S	
3	Covellite	CuS	
4	Malachite	$\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$	57.5
5	Azurite	$2\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$	55.3
6	Chrysocolla	$\text{CuO} \cdot \text{SiO}_2 \cdot 2\text{H}_2\text{O}$	36.2

Analysing the data

- 2 Which of the ores are binary compounds?
- 3 Which ore, as written, could be called a 'hydrate'?
- 4 Why would the presence of $\text{Cu}(\text{OH})_2$ in an ore cause it to be called a basic compound?
- 5 Many ores can have their formulae written in different ways. For example, chalcopyrite, CuFeS_2 , can also be written as $\text{CuS} \cdot \text{FeS}$. Does this alter the proportion of Cu in the ore?
- 6 Which ore could also be written as a type of 'silicate'?
- 7 Calculate the amount (%), by mass, of Cu in the chalcocite and covellite ores. Enter the results in the table.
- a _____
- b _____

Interpreting the data

- 8 Which of the six ores listed contains the greatest proportion, by mass, of Cu?
- 9 Ores are always mixed in with worthless rock called 'gangue'. If malachite was present in a deposit to the extent of 3%, i.e. the gangue made up 97% of the deposit, what amount (%) of Cu would be present?

Data: Set 3 Eliminating the major impurities

After the physical separation of the ore from the gangue, the process begins of separating the copper ore compound from remaining impurities. These impurities are mainly iron oxides and iron sulfides, which are removed by heating to a high temperature with silica (usually silicon dioxide or sand) to form 'slag', a waste material. Two such reactions at this stage are:

- a** Iron(II) oxide + silicon dioxide → iron silicate (slag)
- b** Iron(II) sulfide + oxygen + silicon dioxide
→ iron silicate (slag) + sulfur dioxide

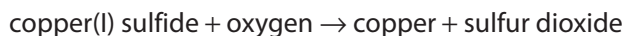
At the high temperature of these reactions the 'slag' is in a molten form and can fairly easily be separated from the desired copper compounds.

Interpreting the data

- 10** Write balanced formula equations for each of the reactions above.
- a** _____
- b** _____
- 11** In what state would the sulfur dioxide probably be as a result of this high temperature reaction?
- _____
- 12** In reaction b it would be important, from an environmental point of view, to collect the sulfur dioxide and not let it escape into the atmosphere. What can happen if this gas does escape into the atmosphere?

Data: Set 4 Extracting the element copper

Once the major impurities have been removed, the remaining copper is present primarily as copper sulfide and contains about 70% copper. Any remaining iron impurities are again treated to remove them as slag. The sulfur in the copper compound is removed at high temperature by blowing hot air into the molten ore compounds to form sulfur dioxide. This can be represented by the equation:



The product of this stage is copper that is about 98% pure. A small amount of impurities remain as oxygen in copper oxide. This can be removed by blowing natural gas through the molten copper oxide. Natural gas can be considered to be mainly methane, CH_4 . This results in a product that is about 99% copper.

Interpreting the data

- 13** Write a balanced equation for the reaction between copper(I) sulfide and oxygen.

- 14** Of all of the copper compounds, it appears that copper oxide is one of the last to break up and release copper metal. What does this suggest about the bond between copper and oxygen in the compound?

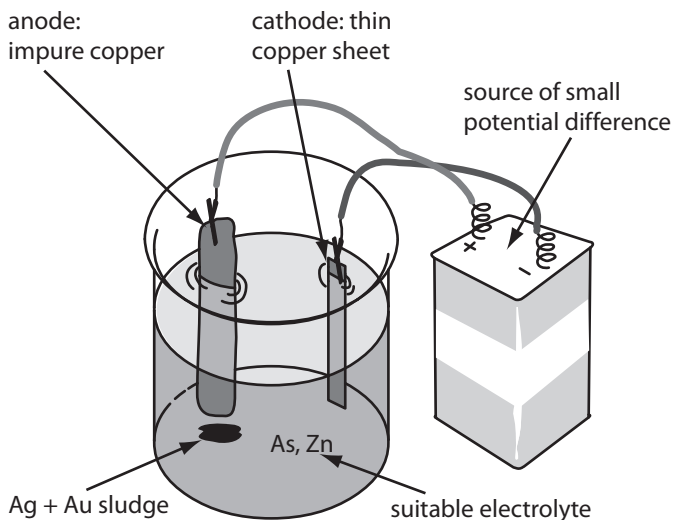
- 15** Would the bond between copper and oxygen most likely be ionic or covalent?

- 16** In the 'final' reaction between residual copper oxide and methane, copper is one of the products. If there are two others, suggest what they might be.

- 17** Using the information in question 16, write a balanced equation for the reaction between copper oxide and methane gas.

Data: Set 5 The final refining stage

To obtain copper with a purity of greater than 99%, an electrolytic process is used. At this stage the copper from the previous processing may contain a few other metals in very small proportions. These may include (1) metals that are more active than copper, such as zinc and arsenic, and (2) metals that are less active than copper, such as silver and gold. The 'impure' copper is used as the anodes in electrolytic cells while thin copper sheets of a very high purity are used for the cathodes. A diagrammatic representation of the working of a single cell is shown below. A suitable electrolyte is used and a small potential difference is applied to the cell.



Interpreting the data

18 Does oxidation or reduction occur at the anode?

19 Write a balanced equation for the reaction involving the impure copper electrode.

20 As the copper anode goes into solution, what would probably happen to the inactive silver and gold atoms as they are released?

21 Copper ions in solution deposit as copper metal on the cathode. What feature of the cell is crucial in preventing zinc and arsenic from being deposited with the copper at the cathode?

22 Write a balanced equation for the reaction at the cathode.

Conclusion

The final copper produced by the above processing is approximately 99.9% pure.

A final interpretation

23 After consideration of all aspects of the five sets of data for this exercise, which of the following four statements would be the most relevant in relation to the energy involved in extracting copper from its ore(s)?

A The process releases more energy than it uses.

B The process consumes a lot of energy.

C The process seems to be energy 'neutral'.

D The process consumes a little bit of energy.

INTRODUCTION

The radiation called infrared radiation (IR) has a frequency range of about 2×10^{13} to 12×10^{13} hertz. Many molecules absorb particular frequencies in this range.

IR spectroscopy is a much-used method for identifying molecules, as a molecule of a particular compound absorbs very specific frequencies. A graph showing these absorbed frequencies becomes a 'fingerprint' of the molecule. To obtain this fingerprint, the compound is vaporised and IR radiation of varying frequencies is passed through the vapour. The amount transmitted is recorded as a graph. A 'dip' in the graph means that IR radiation is being absorbed at that specific frequency. The graphs are commonly referred to as 'IR absorption graphs'. In addition, if a compound is not pure, the impurity absorbs its own frequencies and so can be identified and the amount determined. Hence IR spectroscopy is used to determine the amount of particular compounds in a sample. This enables the purity of the sample to be determined.

Aim

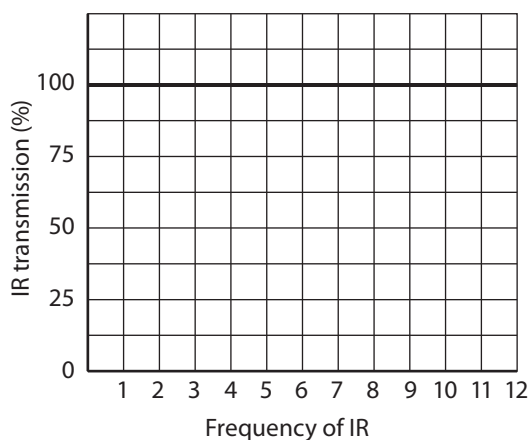
To use the principle of the type of atomic absorption spectroscopy known as 'infrared spectroscopy' to identify some compounds in samples of 'unknowns'.

Data

Some examples to illustrate the process are supplied below. They are simplified and the 'frequency' axis is numbered just to show increasing frequencies rather than specific ones.

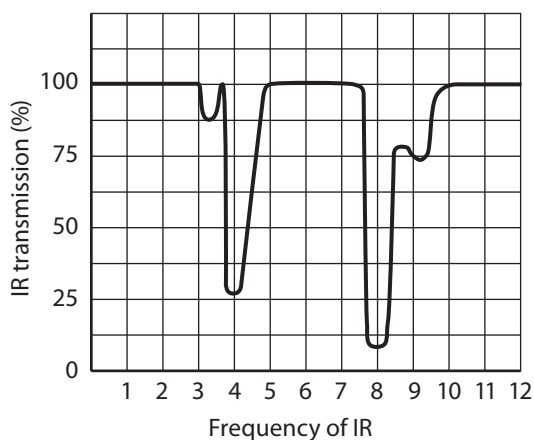
a

Transmission with no sample



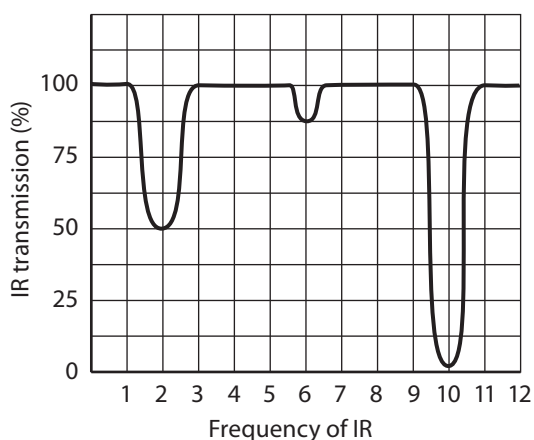
b

IR spectrum of chemical A



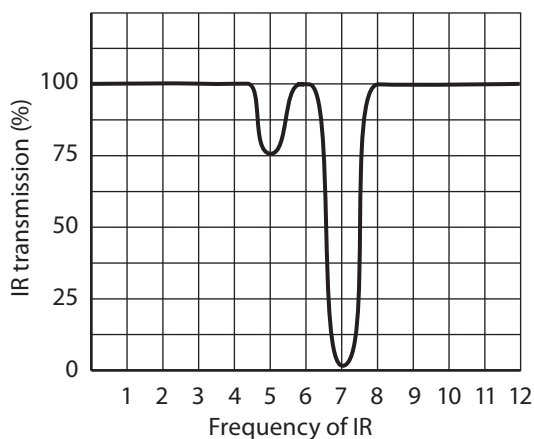
c

IR spectrum of chemical B



d

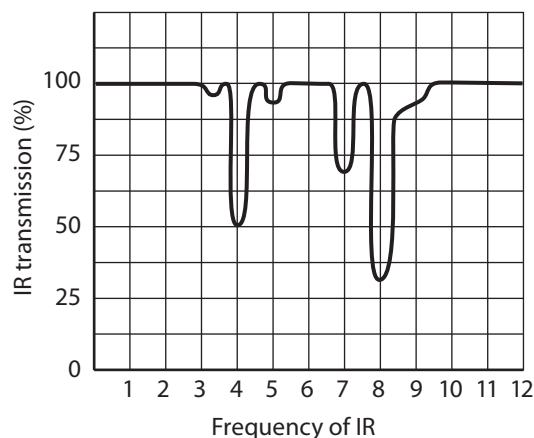
IR spectrum of chemical C



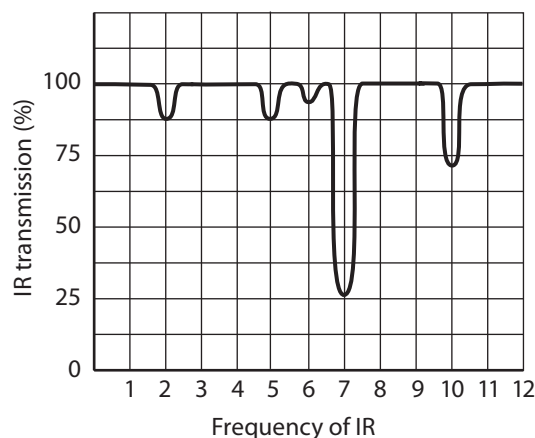
Interpreting the data

The following graphs are those of samples of two 'unknown' compounds, thought to be mixtures of the above chemicals.

a Unknown X



b Unknown Y



1 Identify the compounds in each sample.

X _____

Y _____

2 For each of the unknown samples, identify which of the compounds is the major component.

a X _____

b Y _____

3 When a particular frequency is absorbed and seems to 'disappear', where does the energy go and how does it affect the molecules of the vapour/gas particles?

4 Since the absorbed radiation is a form of energy and is now present in the gas, how does this affect the temperature of the vapour/gas?

5 The reason for the absorption of particular frequencies is that each particular bond between atoms absorbs particular frequencies. For example, C–H bonds and C–C bonds absorb different frequencies and so the frequencies absorbed identify the bonds and hence the elements in them.

In a hydrocarbon such as methane, CH₄, there are only C–H bonds, which all have the same atomic environment. This is not true for C–H bonds in other hydrocarbons such as propane.

a Would it be expected then, that the IR spectrum for propane would be the same as or different from that of methane?

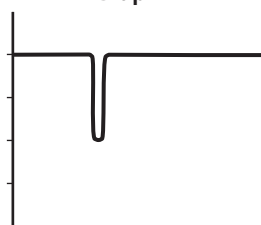
b What other types of bonds are present in propane but are not present in methane?

6 In the two simplified graphs below, one represents the IR absorption graph for propane and the other represents the IR absorption graph for methane. Which graph is more likely to represent:

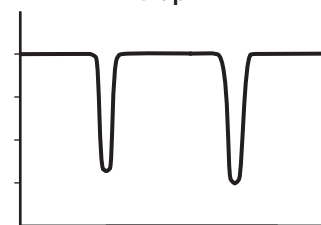
a methane? _____

b propane? _____

Graph 1



Graph 2



7 Propene is another hydrocarbon. Like propane it has three C atoms linked to one another. Is it likely that propene would produce an IR absorption 'fingerprint' that would be the same as the one for propane? (Yes/No) _____

Explain. _____

8 As the C chain in hydrocarbon molecules becomes longer due to the increasing number of C atoms, what would probably happen to the complexity of the IR absorption 'fingerprint'?

9 The microwave radiation is used in a similar way to infrared radiation to identify certain gaseous molecules. The use of microwaves is much more limited than the use of infrared waves.

a Give two ways in which microwaves are similar to infrared waves.

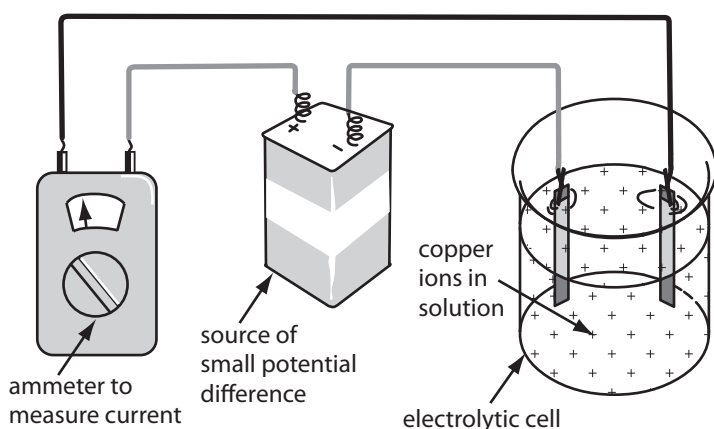
b State one way in which the two types of waves are different.

DATA AND BACKGROUND

Michael Faraday is generally recognised as the principal scientist in the discovery that electricity consists of individual 'units' or is 'quantised'. His electrolysis experiments, carried out long before the suggestion that electrons existed, showed that definite amounts of electricity would deposit definite amounts of metals. His experiments showed that an atom of a metallic element might carry 1, 2, etc. units of charge, but never $1\frac{1}{2}$ or $2\frac{1}{4}$, etc. This led to the eventual acceptance that electricity is composed of particles. Today it is accepted that the particles are negatively charged 'electrons'. This understanding enables predictions to be made about electrolytic cell reactions.

It is now known that 1 coulomb (C) of electrical charge is equivalent in size to 6.25×10^{18} electron charges. If this number of electrons passes through a point in an electrical circuit in 1 second, then a current of 1 ampere (A) is said to flow. The chemist's number, the 'mole', is 6.02×10^{23} and so 1 mole of electrons = 6.02×10^{23} electrons.

An electrolytic cell is set up to deposit copper metal from a solution of its ions.



Aim

To predict how much copper metal will be deposited by a definite amount of electric current passing through an electrolytic cell.

Analysing the data

1 If a current of 2.0 A is allowed to flow through the cell circuit, how many:

a coulombs of charge are delivered in 1 second?

b electrons are delivered in 1 second?

2 How many moles of electrons are delivered in question 1?

3 If the 2.0 A current is allowed to flow for 20 minutes, how many moles of electrons are delivered?

Interpreting the data

4 The equation for the reduction of copper(II) ions is $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu(s)}$. Using the value calculated in question 3, fill in the appropriate spaces in the equation below, indicating moles of reactants and products for this particular reaction.



5 Determine the mass of Cu(s) produced above, given that the atomic mass of copper is 63.5.

6 If the number of moles of electrons supplied to the cathode in the time of the experiment is as calculated in question 3, how many moles of electrons would the power source draw from the anode in the same time?

7 Suppose that a definite mass of copper, say 3.18 g, was required to be deposited in this electrolytic cell. How long would it take to achieve this?

8 In a corresponding silver cell in which Ag^+ ions were being reduced to form Ag(s), the same amount of current is passed through the cell for the same time.

a Would the number of moles of silver deposited in this cell be the same as the number of moles of copper deposited in the original cell? _____

Explain your answer. _____

b What would be the relationship between number of moles of Ag(s) deposited and the number of moles of Cu(s) deposited in the two examples?

c Would this same relationship apply to the masses of the two metals deposited in the reduction processes? (Yes/No) _____

INTRODUCTION

The advent of batteries, particularly small ones, that can be recharged after their charge 'runs out' has been of great benefit in reducing the number discarded after use and ending up in landfill. There is ongoing investigation into improving such batteries.

Aim

To compare the advantages and disadvantages of two commonly used rechargeable battery types.

Data

The two battery types under consideration are (1) the nickel–cadmium battery (NiCd) and (2) the nickel–metal hydride (NiMH) battery. Table 1 summarises many of their features.

Analysing the data

- 1 In Table 1 a comparison is given for the drop in voltage in each battery as it is discharged (i.e. used). Draw a line graph for each battery on the axes provided to show, in a semi-quantitative way, the trends listed.

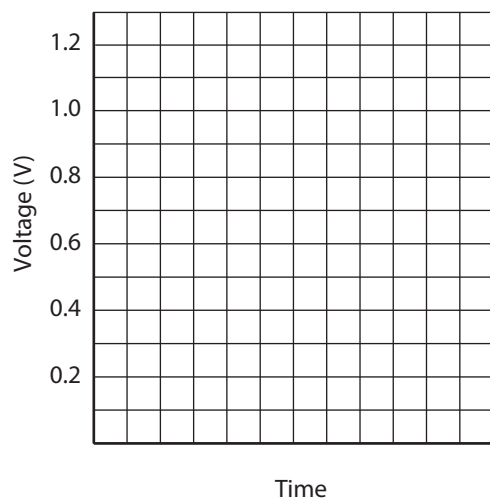


Table 1

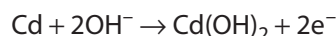
Feature of battery	Nickel–cadmium type	Nickel–metal hydride type
Cell voltage (approx.)	1.2 V	1.2 V
Rate at which charge is lost when not in use (approx.)	10% per month	30% per month (some newer types are improving on this)
Energy stored per unit of volume	50 to 150 units	140 to 300 units
Number of charge cycles during battery life	2000 cycles	500 to 1000
Environmental concerns	• Cadmium is a toxic element. • Nickel can be recycled. • Batteries banned in some countries.	No cadmium, so possibly less environmental impact. Other metals are less toxic.
Change in voltage during discharge	Only a very small drop until discharge is complete.	Voltage drop is more pronounced during discharge.

Electrochemical data

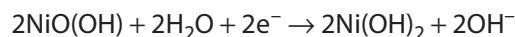
The chemical reactions occurring within these batteries as they produce electricity are given below.

1 The nickel–cadmium type

- a At the cadmium electrode:

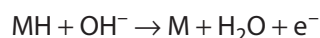


- b At the nickel electrode:

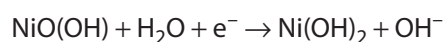


2 The nickel–metal hydride type

- a At the metal hydride electrode:



- b At the nickel electrode:



Interpreting the data

Use the electrochemical data to complete Table 2, which applies to the discharge of the battery.

- 2 According to the electrode reactions given, what type of chemical must be present in the electrolyte of each battery?

- 3 Suggest a suitable, soluble compound for question 2.

Table 2

Indicate	Nickel–cadmium battery	Nickel–metal hydride battery
Electrode undergoing oxidation		
Electrode undergoing reduction		
Which electrode is the cathode		
Which electrode is the anode		
Which electrode is the ‘positive’		
Which electrode is the ‘negative’		

- 4** For each battery, determine the overall net chemical reaction occurring during discharge.

a Nickel–cadmium battery

b Nickel–metal hydride battery

- 5** The E^0 value (i.e. reduction potential) for the electrode reaction involving Ni, which is common to both batteries, is +0.49 V. What would be the E^0 value for the other electrode in:

a the nickel–cadmium battery? _____

b the nickel–metal hydride battery? _____

- 6** The metal in the metal hydride batteries is usually a metal such as lanthanum, cerium, neodymium and praseodymium. What is the name used to refer to these metals in the periodic table?

- 7** Before it can be claimed that NiMH batteries have an environmental advantage because they have no cadmium in them, what other considerations must be taken into account?

- 8** The term ‘battery’ is commonly used to describe these ‘electricity suppliers’. However, this term is, strictly speaking, not correct. Explain why this is so. What is the more correct term?

- 9** For each battery type, list two advantages and two disadvantages as shown in Table 1.

a NiCd type

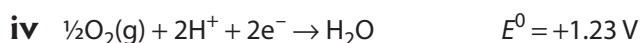
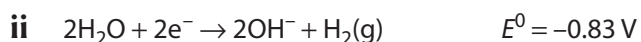
b NiMH type

Aim

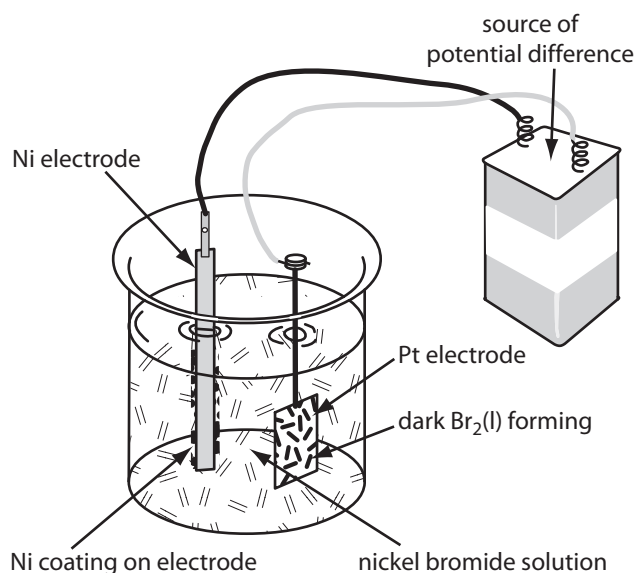
To investigate the reactions that occur in cells that involve a solution of nickel bromide.

Data

Some standard reduction potentials are listed below.



A cell is arranged as shown, using a solution of nickel bromide. A source of potential difference is applied and the resulting current is allowed to flow for some time. After a while some dark $\text{Br}_2(\text{l})$ occurs around the platinum electrode and $\text{Ni}(\text{s})$ is deposited on the Ni electrode.

**Analysing the data**

- On the diagram, show the direction of electron movement in the two connecting wires.
- Complete Table 1 to summarise the various aspects of the cell.

Table 1

Electrode	Made of Pt or Ni	Electrode charge	Half-reaction occurring	Process occurring (oxidation or reduction)	E^0 (V)
Anode					
Cathode					

Table 2

Electrode	Made of Pt or Ni	Electrode charge	Half-reaction occurring	Process occurring (oxidation or reduction)	E^0 (V)
Anode					
Cathode					

- Determine the overall or net reaction occurring in the cell.

- What is the E^0 value for the overall reaction?

- What does the *sign* of the E^0 in question 4 indicate about the spontaneity of the overall reaction?

- What term correctly describes this type of cell?

- If the source of potential difference is now removed and replaced with a voltmeter, what reading could be expected on the voltmeter, assuming correct polarity?

- The cell is now different from the original. Complete Table 2 to show how it is different.

- What term correctly describes this type of cell?

- In which cell (1st or 2nd) do the products gradually gain potential chemical energy as the cell operates? Explain.

- What would gradually happen to the mass of the Ni electrode in the second cell (with the voltmeter) as the cell operates?

Explain.

- If Ni metal were to be extracted from $\text{NiBr}_2(\text{aq})$, which cell would be the one to use?

INTRODUCTION

Catalysts are chemicals that change the rates of chemical reactions while remaining basically unchanged themselves. There is a huge variety of catalysts. They may, for example, be particular metals, acids or metal oxides and they are used extensively in the industrial production of many chemicals. An example is the production of ammonia from nitrogen and hydrogen in the Haber–Bosch process using an iron catalyst.

The search continues to find new and better catalysts that can assist in reducing the amounts of pollutants that enter the air and cause various problems.

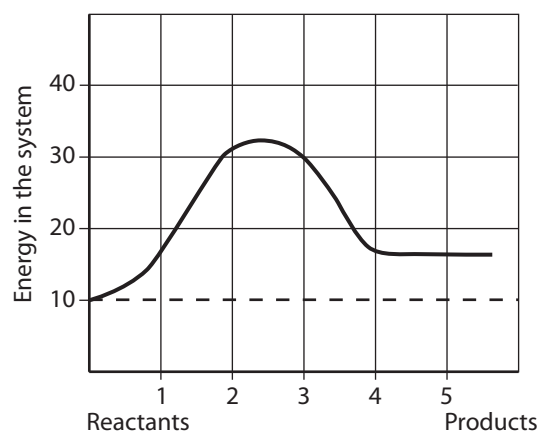
Aim

To investigate the use and effects of some catalysts in situations in which environmental considerations are very important.

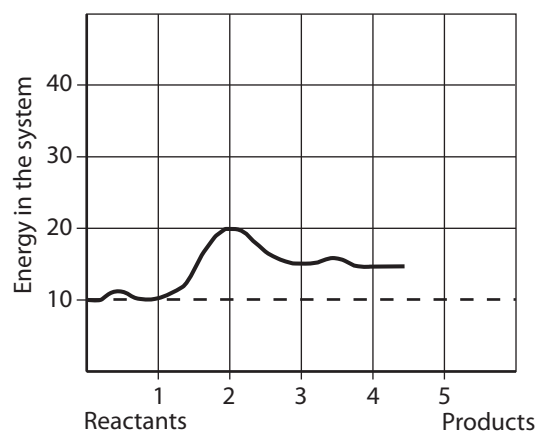
Preliminary aspects

The following graphical representation is that of a catalysed reaction. Two graphs are shown below for a particular chemical reaction. One is for the catalysed reaction, the other for the uncatalysed reaction. These preliminary questions relate to the principle of catalysis in all reactions.

Graph 1



Graph 2



P1 Identify each type of reaction from the graphs.

Graph 1 _____

Graph 2 _____

P2 When the graphs are compared, what differences do they show in relation to:

a time of reaction?

b energy needed to make the reaction proceed?

P3 Do the different pathways for the reaction cause the reactants initially and the products finally to possess more or less energy? Explain.

P4 Is this situation in question P3 the same for all reactions? (Yes/No) _____

Data: Set A Some reactions and their catalysts

	Reactants	Product(s)	Catalyst
a	Sulfur dioxide and oxygen	Sulfur trioxide	Platinum/vanadium
b	Ethene and hydrogen	Ethane	Nickel
c	Nitrogen and hydrogen	Ammonia	Iron
d	Nitrogen dioxide	Nitrogen and oxygen	Platinum/rhodium
e	Carbon monoxide and oxygen	Carbon dioxide	Platinum/palladium

Interpreting the data

1 The metals mentioned as catalysts all belong to a specific group of metals in the periodic table. What is the name of this group?

2 The metals in question 1 differ from the metals of Groups 1 and 2 in the periodic table in their valencies. Without mentioning specific values, what is this fundamental difference?

- 3 Write balanced equations for each of the reactions in Set A. (The states may be omitted.)

a _____

b _____

c _____

d _____

e _____

- 4 Would the use of catalysts in these reactions change the value of ΔH for the reactions?

Data: Set B Car exhausts

Typical car engines that use unleaded petrol burn hydrocarbons with oxygen in air to produce energy. The complete, efficient burning of such fuels produces carbon dioxide and water. If the burning is incomplete, the toxic gas carbon monoxide is formed. Also, due to the high temperature of combustion, a small amount of nitrogen in the air reacts with the oxygen producing nitric oxide gas (NO). This toxic gas can react further with oxygen to produce nitrogen dioxide, another toxic gas and one that starts atmospheric reactions producing photochemical smog, a problem in some major cities. Catalytic converters have, by law, been fitted to all new cars in Australia since 1986. These converters minimise the amount of carbon monoxide and nitrogen oxides in car exhausts.

Interpreting the data

- 5 Use the data in Set B to determine which two reactions in Set A would need to occur in car exhaust systems.

- 6 If nitric oxide breaks down in a catalysed reaction to form the same products as nitrogen dioxide, write a balanced equation to represent this (omit states).

- 7 Car exhaust gases, even after catalysis, contain some carbon monoxide, which readily reacts with oxygen in the atmosphere once they come in contact. This reaction results in the formation of carbon dioxide. Suggest a probable reason for the catalytic converter not being able to complete this conversion.

8 Does the high temperature required to initiate a reaction between nitrogen and oxygen to produce nitric oxide mean that the reaction is automatically endothermic? _____

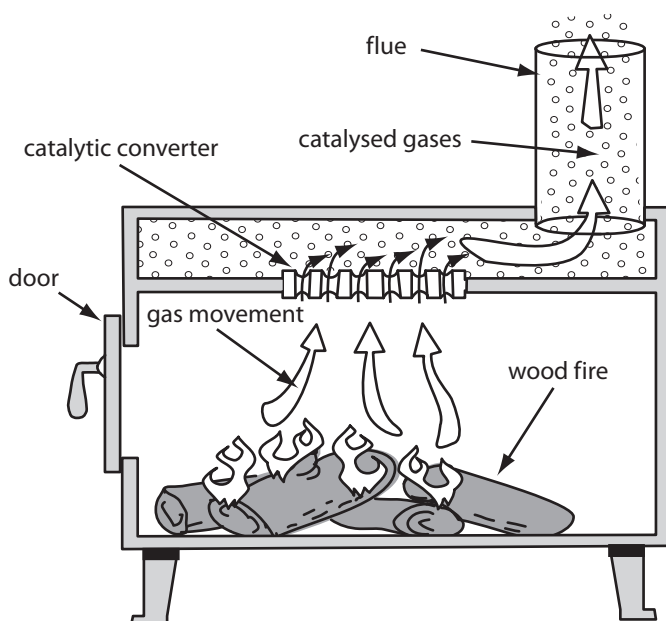
Explain. _____

- 9 The reaction referred to in question 8 has a ΔH value of $+90.4 \text{ kJ mol}^{-1}$ per mole of nitric oxide produced. What does this value represent?

Some additional data

Modern wood-burning heaters used in the home also have catalytic converters that minimise the amount of toxic gases and unburnt particles released into the atmosphere. The temperatures at which these heaters work is much lower than those achieved in a car engine, especially when the fire is 'turned down'. Carbon particles and carbon monoxide are common pollutants. The catalytic converter is installed above the fire. A common catalyst is palladium.

A diagrammatic section through such a heater is shown.



Interpreting the additional data

- 10 Suggest why the catalyst is above the fire.

- 11 Write a balanced equation for the complete combustion of carbon.

- 12 Even when the heater is turned down, the carbon and carbon monoxide are burnt almost completely. Explain how the catalyst allows this to happen.

INTRODUCTION

Most retail products such as fertilisers, hardware chemicals and foods list information about the quantities of important or active ingredients they contain. This can be done in a number of different ways. The following exercise looks at the way information is presented for a particular product.

Aim

To examine the information supplied on a specific product and interpret this in order to gain more detailed information about the product.

Data

Products used to preserve timber and counter the effects of termites and fungi often contain the elements chromium (Cr), copper (Cu) and boron (B). These elements are usually present in compounds that, when mixed with water, can be applied to timber to soak in and so prevent the decay process. One such product, sold in 500 g containers, lists the 'active ingredients' as shown in the table below.

Analysing the data

1 To what general group within the periodic table do the elements chromium and copper belong?

2 The first two compounds in the table are listed as being 'anhydrous'. What does this mean?

3 Determine, from the formulae given, the valency or oxidation number of:

a chromium in the chromate ion.

b boron in the borate ion.

4 Determine the percentage of each element in the compound containing it and enter the values in the table.

5 What is the total mass of effective elements in 1.0 kg of preservative?

6 The instructions say to dissolve 500 g of preservative in 5.0 L of water. Calculate the mass of boric acid that would be in this volume of water.

7 Calculate the approximate molarity of boric acid in the solution.

8 Assuming that boric acid dissociates completely in water, write an equation for this dissociation.

9 What would be the number of moles of H^+ in the 5.0 L of boric acid solution?

10 Express the concentration of H^+ ions from question 9 in mol L^{-1} .

11 Copper sulfate, when in the 'hydrous' or hydrated form, has the formula $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$. Would the amount (%) of copper in the anhydrous form be the same as in the hydrous form?

Explain.

12 Sodium compounds such as NaCl , Na_2SO_4 and Na_2CO_3 are colourless but the preservative sodium chromate, $\text{Na}_2\text{Cr}_2\text{O}_7$, has a pronounced orange colour. Which element might be the source of the colour?

Atomic masses: H = 1, B = 11, O = 16, Na = 23, S = 32, Cr = 52, Cu = 63.5

No.	Preserving element	Compound containing element	Amount of element present in 1.0 kg preservative	% of element in compound
1	Chromium (Cr)	Anhydrous sodium chromate ($\text{Na}_2\text{Cr}_2\text{O}_7$)	160.0 g	
2	Copper (Cu)	Anhydrous copper(II) sulfate (CuSO_4)	115.0 g	
3	Boron (B)	Boric acid (H_3BO_3)	50.0 g	

INTRODUCTION

The greyish powder that can be purchased from hardware stores and used, commonly, for making 'concrete' is correctly called 'Portland cement'. It has been used for more than 150 years. The chemistry of cement is quite complex. Its production involves many chemicals; some simple, some complex.

Aim

To look at aspects of the chemistry involved in the production of what is commonly called 'cement'. Important aspects considered here are (1) the ingredients, (2) the energy needs and (3) the environmental concerns.

Data: Set 1 Ingredients

These usually come from local rocks that contain important minerals that are compounds such as oxides of calcium, magnesium, aluminium, iron and silicon. The different rock sources vary in their mineral content. To keep the composition of the cement powder constant, other minerals are added to compensate for those deficient in the source or local rock. The mixture is eventually crushed to a very fine powder after some initial chemical processing. The most common source rock is 'limestone', usually an impure one. Pure limestone is calcium carbonate.

Analysing the data

- I Determine the formulae for the oxides of:
 - a calcium. _____
 - b magnesium. _____
 - c aluminium. _____
 - d iron(II). _____
 - e iron(III). _____
- 2 The formulae for some compounds can be written as 'combinations' of the oxides from which the compound may be made. For example, CaCO_3 can be represented as $\text{CaO} \cdot \text{CO}_2$ and $\text{Mg}(\text{OH})_2$ as $\text{MgO} \cdot \text{H}_2\text{O}$. Using this principle, complete the following table.

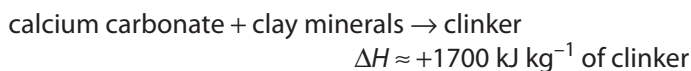
No.	Formula	Formula written as oxides	Name of compound
1	CaSiO_3		
2	Al_2SiO_5		
3	MgAl_2O_4		
4	Fe_2SiO_5		
5	Fe_2SiO_4		

- 3 What is the molar ratio of MgO to Al_2O_3 in example 3 in the table?

- 4 Is the ratio for question 3 the same for the oxides in the other examples in the table?

Data: Set 2 Energy considerations

The ingredients are heated in a rotating kiln and subjected to increasing temperatures up to a peak of about 1400°C . At this stage there is partial melting of the ingredients and many chemical reactions have occurred. On cooling, the resulting product is a coarse-grained solid called 'clinker'. The reactions that have occurred have produced a lot of carbon dioxide. An approximate representation, in words, for the reactions is:



Clinker is later ground in a mill and calcium sulfate is added to make a final powder or 'cement'. A large grinding mill may process, for example, 200 tonnes of clinker per hour to make cement.

Interpreting the data

- 5 The ΔH value for the clinker reaction has a '+' sign. What does this mean about the reaction?
- 6 Calculate the ΔH for a mill that processes 200 tonnes of clinker in an hour.

Data: Set 3 Environmental considerations

The manufacture of cement starts from the time the raw materials are extracted from the ground and ends when the product is packaged for sale.

Interpreting the data

- 7 Considering the ΔH value given and the high temperature required to produce cement, make a statement about the energy involved in its manufacture.
- 8 Consider the process that occurs in the production of cement from the extraction of the original rock ingredients to the final powdered product. What are some environmental effects that are of concern?

INTRODUCTION

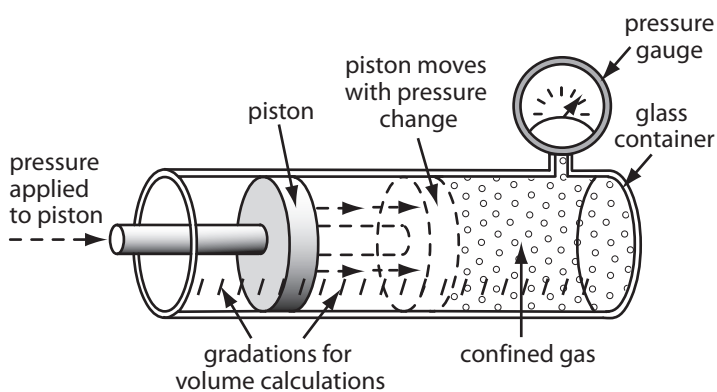
Robert Boyle, in the 17th century, investigated the effect of increasing pressure on the volume of a fixed mass of gas that was maintained at a constant temperature. This was long before atomic theory was established. This exercise repeats, in principle, the work done by Boyle.

Aim

To examine and interpret some data from a modern experiment that investigates the effect of increasing pressure on a fixed mass of gas at constant temperature.

Data

The diagram shows the basic equipment used. The mass of gas in the experiment was 0.4 g and the temperature was constant at 25°C (298 K). The original pressure on the gas was standard pressure of 101.3 kPa (1 atm) and this was gradually increased throughout the experiment. The results are listed in the table below.



Note: These results are theoretical.

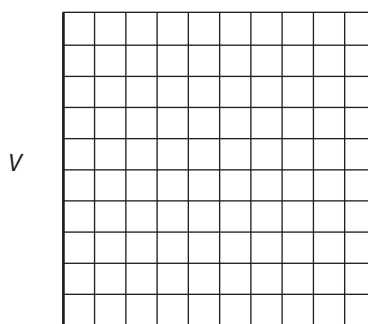
No.	Pressure applied (kPa)	Resulting volume (L)	$\frac{1}{\text{Volume}}$	Pressure $\times$ volume
1	101.3	2.45		
2	152.0	1.63		
3	202.6	1.23		
4	253.3	0.98		
5	303.9	0.82		
6	358.3	0.69		

Analysing the data

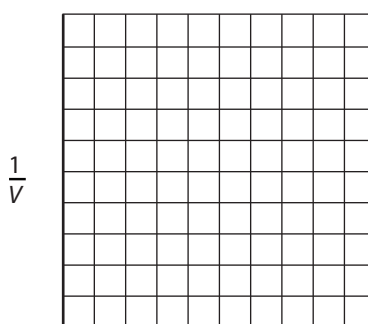
- 1 Calculate the values for the remaining two columns and enter them appropriately.
- 2 Construct three graphs from the data:
 - a Graph 1: Volume vs Pressure
 - b Graph 2: $\frac{1}{\text{Volume}}$ vs Pressure
 - c Graph 3: Pressure $\times$ volume vs Pressure

Use the units given in the table.

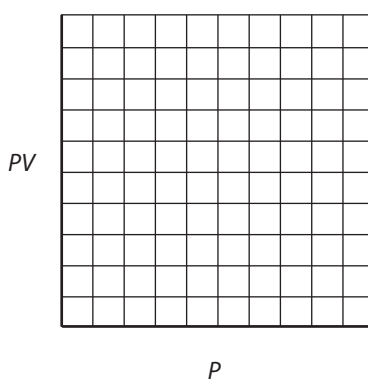
Graph 1



Graph 2



Graph 3



Interpreting the data

Note: A periodic table may be needed here.

- 3 Why has 'pressure' been plotted on the x-axis of each graph?
- 4 What word describes the shape of graph 1?
- 5 What does the shape of graph 1 suggest about the relationship between P and V ?
- 6 Does the shape of graph 2 support the suggestion about graph 1?

7 Use your graphs to predict:

a the pressure needed to cause the gas to have a volume of 1.1 L.

b the volume of the gas when the pressure on it is 325 kPa.

8 Given that one mole of any gas at 101.3 kPa pressure and at 298 K occupies 24.5 L, what fraction of a mole of gas has been used in this experiment?

9 a What must be the molar mass of the gas?

b Suggest what the gas might be.

10 Make a statement in words to describe the findings as shown by the graphs.

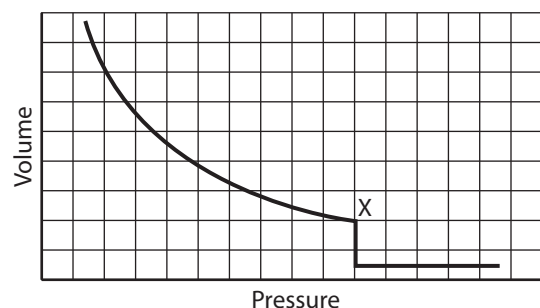
11 Write a mathematical expression for the relationship between pressure and volume as revealed by graphs 1 and 2.

12 What does graph 3 reveal about the product $P \times V$ even as the pressure is increased?

13 Would the general shape of graph 1 be different if 4.0 g of the same gas had been used?

Data for a different gas

Although the above findings apply in principle to all gases, the relationship revealed in these graphs applies only to a specific range of temperatures for each gas. The experiment was repeated with a different, heavier gas and with higher pressures, and the resulting data plotted in the graph below.



Interpreting the graph

14 Suggest a reason for the sudden drop in volume at point X.

15 Does this new shape mean that the relationship previously found does not apply at all for this gas?

Explain.

16 What does this sudden change of shape indicate about trying to extrapolate (outside the measured range of the data) in graph 1?

17 The shape of the graph to the right of X remains a straight line parallel to the x-axis even though the pressure is increasing. Explain why this happens.

18 When a gas is compressed as in these experiments, the molecules of the gas lose kinetic energy. Where does this energy go?

19 Considering the answer from question 18, what problem would this create in carrying out any such experiment at constant temperature?

Conclusion

20 The discovery by Robert Boyle as shown by the original graphs 1, 2 and 3 have been expressed as 'Boyle's law'. Make a statement to express this law in your own words.

21 Why is Boyle's discovery called a law and not, for example, a theory?

INTRODUCTION

A much sought after chemical compound in the chemical industry is sodium carbonate, commonly called 'soda ash'. It is used in many industrial processes such as making glass, soaps and detergents and in water treatment. The worldwide production of soda ash is enormous and the most common method of producing it is by the 'Solvay process', an ingenious recycling process that uses easily obtainable ingredients. It has been used since the 1860s.

Aim

To investigate and analyse the principle chemical reactions that occur in the Solvay process.

Data

The reactions in this process are complex, but a simplified version is considered here by dealing with four basic steps and the fundamental chemical reactions involved, as set out in the box below.

Analysing the data

I Write balanced equations for:

a the overall net reaction.

b the reactions involved in the four basic steps.

A: _____

B: _____

C: _____

D: _____

Interpreting the data

2 Which two reactions are obviously endothermic?

3 By studying the four equations, list those chemicals that can be cycled back in earlier reactions.

4 Suppose that 1 mole of sodium chloride is used in reaction A. How many moles of:

a ammonia are required?

b sodium hydrogen carbonate are produced?

c ammonium chloride are produced?

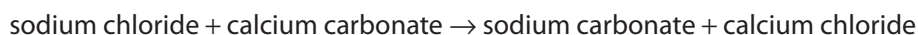
5 This question assumes that no loss occurs when any chemical is recycled. According to the balanced equations, would the amount of ammonia produced in reaction C be enough to cause the complete reaction with another mole of sodium chloride in reaction A? (Yes/No)

6 Suggest an easily obtainable source for:

a the sodium chloride solution.

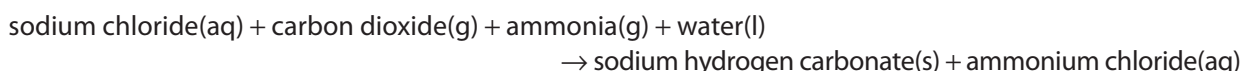
b the calcium carbonate solid.

The overall net reaction in the Solvay process can be represented by the following word equation:

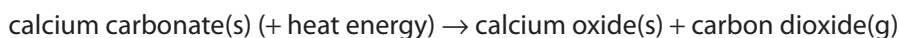


The four basic steps

Reaction A:

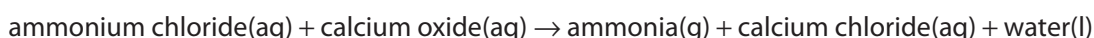


Reaction B:

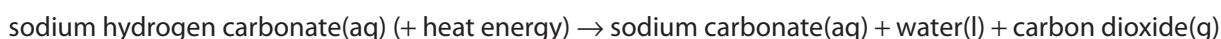


Note: This reaction occurs at about 1000°C.

Reaction C:



Reaction D:



Note: This reaction occurs at about 200°C.

INTRODUCTION

The term 'desalination' used in its simplest context means 'to take the salt out of a solution'. This process is often presented as a means to provide enough drinkable (potable) water for human populations where the population density is high. The obvious source is the saline water of the oceans. Desalination has been used for some time by Middle Eastern countries and many island communities, and has been proposed for Australian coastal cities where the populations are dense. It involves a process called 'reverse osmosis'.

Aim

To investigate the chemistry of the desalination process and some of the consequences of it.

Data: Set 1 Osmosis

To understand 'reverse osmosis', the process of normal 'osmosis' is best dealt with first. Osmosis is the passage of water through a semipermeable membrane to equalise the concentrations of solutions on each side of the membrane. The dissolved salts themselves don't move through the membrane. An example is shown in Figure 1 in which a semipermeable membrane separates two solutions of unequal concentrations.

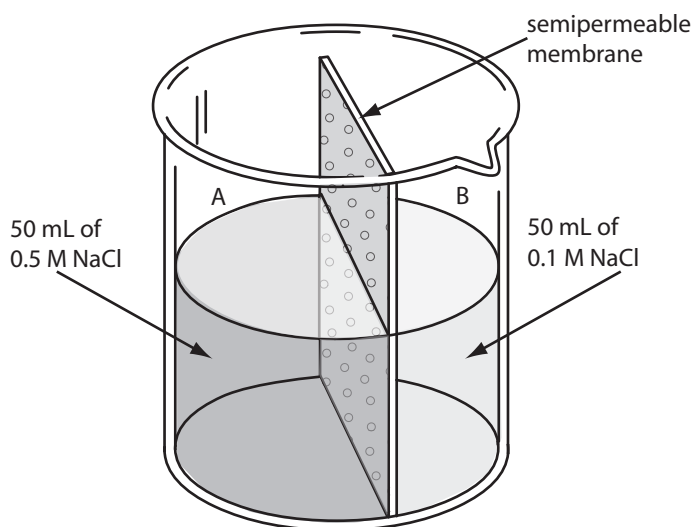


Figure 1

The dissolved salt here is sodium chloride and the solution on side A of the membrane is more concentrated than that on side B.

Analysing the data

1 In which direction would water have to move through the membrane such that the solutions on side A and side B could achieve equal concentrations?

A Side A to side B

B Side B to side A

2 As this movement occurs, what happens to the concentration of the solution on:

a side A? _____

b side B? _____

3 Determine the number of moles of solute in each solution before water movement takes place.

Solution A _____

Solution B _____

4 Would the values in question 3 alter as osmosis proceeds? (Yes/No) _____

5 Assume that x mL of water moves through the membrane to achieve equal concentrations on both sides. Write a simple expression for the volume of solution on each side.

a Side A _____

b Side B _____

6 Using information from above, calculate:

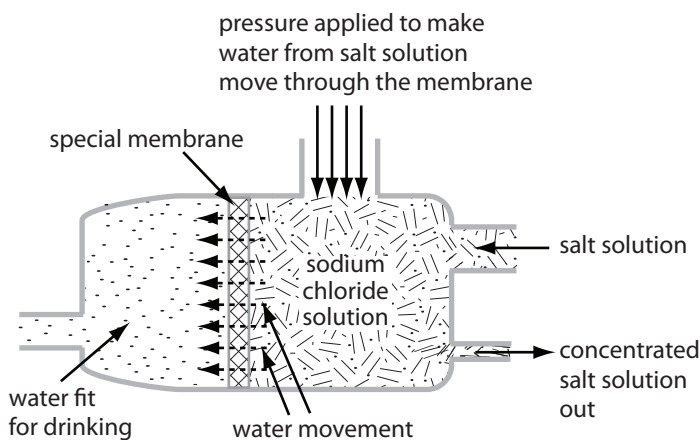
a the final concentration of the two solutions,

b the volume of water that moves through the membrane, and hence

c the volume of solution on each side of the membrane.

Data: Set 2 Reverse osmosis

In 'reverse osmosis' it is the water that is forced through special membranes which prevent the flow of sodium and chloride ions. Hence the water after it has passed through the membranes is relatively pure and fit for drinking. The water remaining on the other side of the membrane is returned to the sea. The process is represented diagrammatically in Figure 2.

**Figure 2****Additional data**

Note: The density of sea water is a little greater than that of pure water but the value of 1.0 kg L^{-1} (1.0 g mL^{-1}) is used here. The more solute present, the greater the density of the solution.

Interpreting the data

- 7** The approximate content of sodium chloride in sea water is 3.5% by mass. Calculate the molar concentration of NaCl in sea water.

- 8** If, for example, all of the NaCl in 1 L of sea water could be extracted and added to a second litre of sea water, what would be [NaCl] in the second litre?

- 9** In desalination plants near tidal estuaries, brackish waters at the end of the outgoing tide in rivers are preferred for treatment in the plants. Suggest why this is the case.

- 10** A particular plant claims that 85 L of drinkable water is produced for every 100 L of water processed, leaving 15 L of concentrated brine (NaCl solution). At 3.5% concentration by mass, what mass of NaCl would be present in 100 L of sea water?

- 11** Determine the concentration of NaCl in the 15 L in:

a g L^{-1} . _____

b mol L^{-1} . _____

- 12** How many times more concentrated is this solution than normal sea water?

- 13** Since the density of the 15 L of concentrated brine is greater than that of normal sea water, what would the concentrated brine tend to do because of its density, if returned to the sea without dilution?

- 14** Considering that marine plant life and animal life have evolved with a relatively constant amount of salt in the oceans, what could be the effect on the marine life of introducing the concentrated brine?

- 15** If a proposed desalination plant at Kurnell, near Sydney, NSW, goes into full production, it is expected to produce $500 \times 10^6 \text{ L}$ per day of potable water. It also means that an enormous amount of concentrated brine will be returned to the ocean daily. What will have to be done to this brine as it is returned to ensure its effects on marine life are negated or at least kept to a minimum?

INTRODUCTION

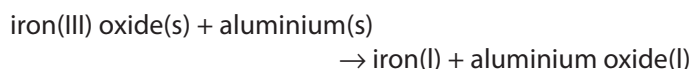
An interesting and useful chemical reaction is the one called the 'thermite' reaction. As its name suggests, the reaction involves the release of a great deal of heat energy. It is this energy that makes the reaction so useful. For example, the heat energy can cause the melting of iron or steel and so the reaction can be used for welding metals.

Aim

To look at the different chemical aspects of the thermite reaction.

Data: Set 1

A thermite mixture is usually a powder containing an active metal and a less active metal oxide. The mixture is quite safe to handle, even though the ingredients can react very vigorously. One reaction that is regularly used is:



$$\Delta H = -853 \text{ kJ mol}^{-1} \text{ per mole of iron(III) oxide used}$$

Note: The products are liquid because of the high temperatures produced.

Atomic masses: Fe = 56, O = 16, Al = 27

To ignite the reactants, a 'primer reaction' is used as a 'fuse'. This reaction itself is strongly exothermic. One primer reaction that is used is the burning of magnesium in air.

Analysing the data

- Write a balanced equation for the thermite reaction and include the relevant ΔH value.

- Write a balanced equation for the 'primer reaction', showing the sign of ΔH .

Interpreting the data

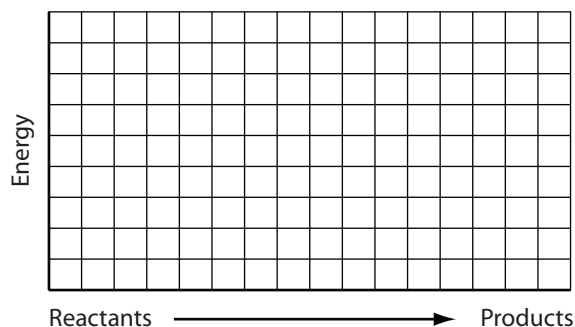
- The thermite reaction is really a 'redox' reaction. Identify which chemical is:

- oxidised. _____
- reduced. _____

- Write half-equations for:
 - the oxidation reaction. _____
 - the reduction reaction. _____

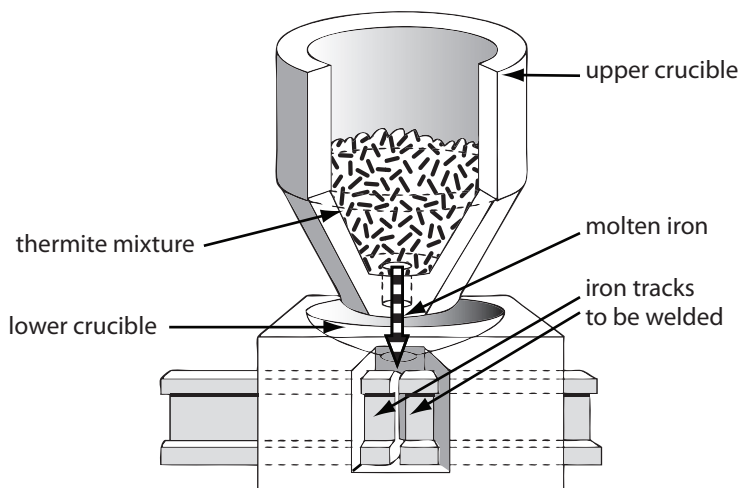
- The powdered thermite mixture can be safely transported over long distances to sites where it is to be used. What does this suggest about the activation energy for the thermite reaction?

- Considering the energy changes as the reaction proceeds, draw an approximate energy diagram for the thermite reaction on the axes provided. Mark in the section that represents ΔH for the reaction.



Data: Set 2

The thermite reaction is used to weld (fuse together) rail and tram tracks. To do this, the following set-up is used. The reaction may reach a temperature of 2500°C, well above the 1530°C needed to melt iron. The molten iron from the reaction in the upper crucible flows into the lower crucible and 'welds' the iron rails or tram tracks together.



Interpreting the data

7 As the high-temperature molten iron from the upper crucible surrounds the parts to be welded, what would physically happen to these parts?

8 A package of thermite mixture is marked '40 kg'. Assuming that the reactants are in the exact proportions for a reaction, determine the mass of each reactant present in the package.

9 From the balanced equation for the reaction, calculate the heat energy produced by a 40 kg package as in question 8.

Data: Set 3

The ΔH value of the overall thermite reaction can be checked by using the values of ΔH for the formation of the oxides of iron and aluminium from their elements. If one mole of each of Al_2O_3 and Fe_2O_3 are formed from the elements of which they are composed, the ΔH values for such reactions are:

**Interpreting the data**

The two reactions can be manipulated to produce the thermite reaction. Whatever is done to the equation is also done to the ΔH value.

10 What needs to be done to each of the equations to produce the thermite reaction?

a R_1 _____

b R_2 _____

c And then _____

11 Apply the above procedures to the ΔH values and so determine the final ΔH value.

12 How does this value compare with the original value given for the thermite reaction?

13 Metals such as Zn and Pb have oxides that would also react with aluminium, but they are never used in the thermite reaction. Suggest a reason for this.

INTRODUCTION

The chemical reaction known as photosynthesis is perhaps the most important reaction that occurs on Earth. Without it life as we know it would not exist. The reaction is carried out by green plants and involves the conversion of carbon dioxide and water into simple sugars such as glucose. Oxygen is also a product of the reaction, which needs to absorb the Sun's energy to proceed. Chlorophyll is a chemical in the leaves of plants, which absorbs the Sun's energy and allows this complex process to occur.

Aim

To look at some of the chemical aspects of the photosynthetic process.

Data: Set 1 Molar considerations

The photosynthesis reaction can be represented by a balanced chemical equation. The word equation is:



Relevant atomic masses: C = 12, H = 1, O = 16

Interpreting the data

- 1 Write a balanced chemical equation for the photosynthesis reaction.

- 2 Determine the molar mass of glucose.

- 3 What mass of carbon dioxide is needed to produce 1 mole of glucose?

- 4 If 1 mole of carbon dioxide gas occupies 24.5 L at normal temperature and pressure, what volume of the gas is needed to form 1 mole of glucose?

- 5 An average adult is said to produce about 1 kg of carbon dioxide per day due to normal respiration and breathing. How many moles of carbon dioxide does this represent?

- 6 If the carbon dioxide in question 5 is converted to glucose by the photosynthesis reaction, how many moles of glucose are formed?

- 7 What mass of oxygen gas would be produced in question 6?

Data: Set 2 Energy considerations

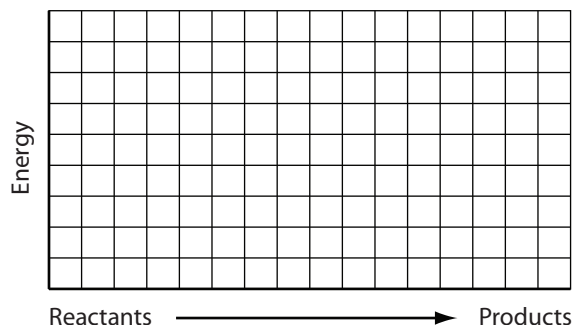
The balanced equation for the photosynthesis reaction has a ΔH value of approximately +2830 kJ per mole of glucose produced. This energy comes from the Sun and is stored in the glucose molecules.

Interpreting the data

- 8 Write the full photosynthesis equation and include the ΔH term.

- 9 Is the photosynthesis reaction exothermic or endothermic?

- 10 Sketch an energy graph to show the difference between the energy of the reactants and products as the reaction proceeds. Mark in the section that represents the ΔH value.



- 11 The reaction for respiration is the reverse of that for photosynthesis.
 - a What is the sign of ΔH for this reaction?

 - b Is the respiration reaction exothermic or endothermic?

INTRODUCTION

In nature, most elements occur in a combined state with other elements. The compounds of metals are called *ores*, and there are many different processes used to extract the metal from its ore. The process chosen for a particular metal depends on a number of factors including the activity of the metal and the cost of the process.

Aim

To investigate some of the processes and the chemistry involved in the extraction of a few metallic elements from their ores.

Data: Set 1 Pure elements

A few elements, both metals and non-metals, do occur in nature as the pure element. These are fairly obvious.

Interpreting the data

- 1** Give the symbols of **a** four metallic and **b** four non-metallic elements that occur in nature in the elemental state.

a _____

b _____

- 2** If an element occurs in its pure state, what does this mean about its ability to react with commonly occurring substances such as water, oxygen and carbon dioxide?

Data: Set 2 Processes

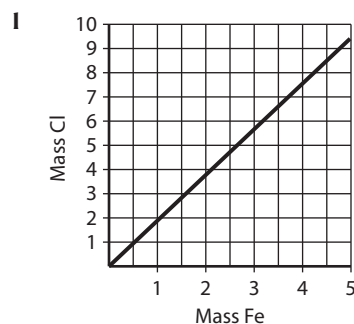
Some commonly used extraction processes are listed.

- a** Reduction of the metallic oxide by heat: Usually a reductant is used to assist the process. Examples include the extraction of iron from iron oxide using carbon monoxide, and the extraction of manganese from the ore pyrolusite (manganese dioxide) using aluminium.
- b** Conversion of a sulfide ore to its oxide by 'roasting' in air and the subsequent reduction of the oxide to the metal by a suitable reductant: Examples here are the roasting of zinc blende (zinc sulfide) and galena (lead sulfide) to form oxides and their reduction to the metal by the reductant carbon.
- c** Electrolytic reduction of molten halides of the metals: Examples include the electrolytic reduction of sodium from rock salt (sodium chloride) and potassium from a chloride ore.

Interpreting the data

- 3** In the reduction of haematite (iron(III) oxide), carbon monoxide is used as the reductant. Write a balanced equation for this reaction.
- _____
- 4** In the reduction of pyrolusite (manganese(IV) oxide), aluminium is used as the reductant. Aluminium oxide and manganese are formed. Write a balanced equation for this reaction.
- _____
- 5** In such a process, what happens to the reductant during the reaction?
- _____
- 6** For sulfide ores such as zinc sulfide and lead sulfide, the preferred treatment is to 'roast' the ore in air and then convert it to the oxide and sulfur dioxide. Give balanced equations for the reaction of:
- a** zinc sulfide. _____
- b** lead sulfide. _____
- 7** The metal oxides produced in question 6 are then reduced to the metals by reacting them with carbon. Suggest a reason why this 'two-step' method is preferred to simply reducing the sulfide ores to their metals.
- _____
- _____
- 8** Sodium metal cannot be extracted from rock salt by either of the reducing methods used above. This is also the case for potassium metal. Suggest a reason for this.
- _____
- _____
- 9** When fused rock salt is electrolysed:
- a** to which electrode – the anode or the cathode – would the sodium ions be attracted?
- _____
- b** which process, oxidation or reduction, would these ions undergo to produce sodium metal?
- _____
- 10** Why is it not possible to electrolyse an aqueous solution of sodium chloride to produce sodium metal?
- _____
- _____

1 Mass analysis in the formation of a binary compound



- 2 1.90:1, 1.90:1, 1.91:1, 1.90:1, 1.90:1 3 Straight line
4 Constant at 1.90:1 5 Yes
6 a 7.6 g b 1.05 g
7 34.4% 8 Yes
9 Atomic mass of Cl and Fe 10 Option 2

2 Mass analysis in a chemical reaction

- 1 $\text{CuO(s)} + \text{H}_2\text{(g)} \rightarrow \text{Cu(s)} + \text{H}_2\text{O(g)}$
2 a Copper b Water
3 2.400 g 4 2.702 g
5 Oxygen is lost from copper oxide.
6 Water is formed in the reaction.
7 All water produced must have been absorbed by tube B.
8 It acts as a check that all water has been absorbed by tube B.
9 The mass of hydrogen is not considered.
10 Endothermic 11 a Copper oxide b Hydrogen
12 Yes. New chemicals often involve a colour change.

3 Empirical and molecular formulae of a compound

- 1 Benzene 2 No
3 a 2.88 g b 0.24 g
4 3.12 g 5 It is the same.
6 Yes 7 C and H
8 a 92.3 b 7.7
9 C:H = 7.7:7.7 = 1:1 10 CH
11 13 amu 12 78/13 = 6 13 C_6H_6
14 C and H 15 CH 16 C_6H_6
17 Structural formula 18 Spatial arrangement of atoms

4 Water of crystallisation

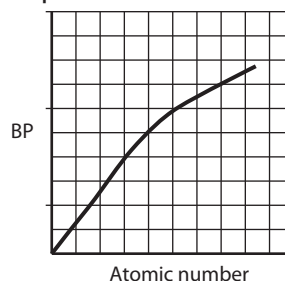
- 1 a 208 g b 106 g c 159.5 g d 130 g
2 a Sodium carbonate b Sodium carbonate
3 a 36 g b 180 g c 90 g d 108 g
4 a 14.8 b 62.9 c 36.1 d 45.4
5 Sodium carbonate 6 100°C
7 Mass of salt used must include mass of water of hydration.
8 19.7 g 9 0.37 M
10 Heat the salt to constant weight before weighing out 53.0 g.

5 Noble gases

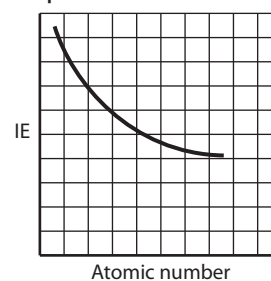
Table 2

No.	NG	AN	AM	MM	MP	BP	IE $\times 10^2$	No. of isotopes
1	He	2	4	4	-272	-269	23.6	2
2	Ne	10	20	20	-249	-246	20.1	3
3	Ar	18	40	40	-189	-186	15.2	3
4	Kr	36	84	84	-157	-152	13.5	5
5	Xe	54	131	131	-112	-108	11.7	9
6	Rn	86	222	222	-71	-62	10.3	0

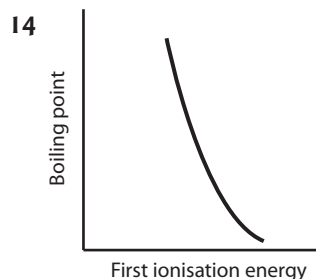
- 1 Table 2 2 Extremely low
3 -62°C 4 Very high
5 Graph A



Graph B



- 6 a Graph A. As atomic number increases so does BP.
b Graph B. As atomic number increases IE decreases.
7 High: atom of element not likely to react by losing electrons.
Low: atoms more likely to react by losing electrons.
8 No. Other radioactive elements (e.g. U) have isotopes.
9 Hydrogen, oxygen, nitrogen, fluorine, chlorine
10 Other gases form stable diatomic molecules.
11 Helium 12 Helium
13 Easier to remove electrons from xenon atoms and so make it combine.



- 15 Atoms of gases have complete outer electron shells and so won't receive, give or share electrons. Hence no combination.
16 Inert gases

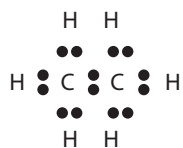
6 Organic chemistry I: Hydrocarbons with two carbon atoms

- 1 Decreases by 2 (-ane to -yne).
2 Decreases by 2.
3 No. All very similar.
4 Yes, the C-C bond length decreases.
5 Fewer H atoms per molecule
6 Decrease because bond become stronger: single to double to triple bonds.
7 C-H bonds are all of a similar type, i.e. single bonds.

8 As bond lengths decrease, bond energies increase.

9 Ethyne

10 a Ethane



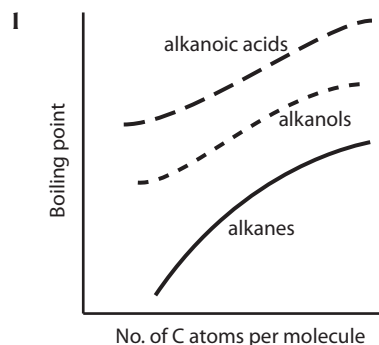
b Ethene



c Ethyne



7 Organic chemistry II: Boiling points of some organic compounds



2 Homologous series

3 a C_nH_{2n+2} b $C_nH_{2n+1}OH$ c $C_{n-1}H_{2n-1}COOH$

4 As the number of C atoms increases so does the BP.

5 a Yes b Yes

6 All are liquids except methane, ethane, propane and butane, which are gases.

7 It is transported at high pressure.

8 They are very combustible in air and have low ignition temperatures.

9 More intermolecular forces occur as molecules become more complicated.

8 Gases in the atmosphere

1 $O_2 = 16$, $N_2 = 14$, $CO_2 = 22$, $Ar = 20$, $Ne = 10$

2 Passes directly from solid to gas with increase in temperature without the intermediate liquid state.

3 Reacts with water to make new products.

4 a CO_2 b Ne 5 3; CO_2 , O_2 , Ar

6 CO_2 sublimates, hence no 'wet' or liquid state.

7 Bring the mixture into contact with $Ca(OH)_2(aq)$ (lime water); white colour indicates CO_2 .

8 Ne, N_2 , O_2 , Ar, CO_2

9 Easiest to extract temperature-wise and doesn't support combustion. Also most dense and 'sits' on fire.

9 The atmosphere and jet aircraft

1 $C_{10}H_{22} + \frac{31}{2}O_2 \rightarrow 10CO_2 + 11H_2O$

2 730 g 3 142 g 4 5.14

5 228 420 L 6 1.667×10^8 g 7 5.824×10^8 g

8 5.165×10^8 g 9 2.63×10^8 L

10 Lower temperature and pressure at high altitude.

10 Beach sand

1 Coal is a mixture.

2 Silicon dioxide, calcium carbonate, titanium oxide, iron titanium oxide, zirconium silicate

3 Insoluble for all.

4 No. They are all insoluble.

5 Reaction with HCl

6 a 4 b 4 c 2 d 4

7 Calcite and quartz

8 Yes. Coal, calcite and quartz would float, others sink.

9 Shells of marine organisms. 10 60% 11 3.0 g

11 Coal

1 a $C + O_2 \rightarrow CO_2$

b $H_2 + \frac{1}{2}O_2 \rightarrow H_2O$

c $N_2 + 2O_2 \rightarrow 2NO_2$

d $S + O_2 \rightarrow SO_2$

2 a 93.1 g b 2.4 g c 1.0 g d 0.8 g

3 a 341.4 g b 1.6 g c 3.3 g

4 251.4 g 5 6.8×10^{13} g

6 a decrease b-d increase

7 acidic 8 a anthracite b lignite

12 Heat of combustion

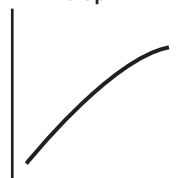
1 Molar masses (g): $CH_4 = 16$, $C_2H_6 = 30$, $C_3H_8 = 44$, $C_4H_{10} = 58$, $C_5H_{12} = 72$

ΔH_c (kJ g⁻¹): $CH_4 = 55.6$, $C_2H_6 = 52.0$, $C_3H_8 = 50.5$, $C_4H_{10} = 49.6$, $C_5H_{12} = 48.7$

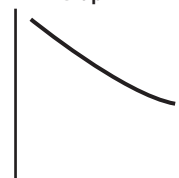
2 Table 2

1	$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$	64	44	4.00	2.75
2	$C_2H_6 + \frac{7}{2}O_2 \rightarrow 2CO_2 + 3H_2O$	112	88	3.73	2.93
3	$C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$	160	132	3.63	3.00
4	$C_4H_{10} + \frac{13}{2}O_2 \rightarrow 4CO_2 + 5H_2O$	208	176	3.59	3.03
5	$C_5H_{12} + 8O_2 \rightarrow 5CO_2 + 6H_2O$	256	220	3.56	3.06

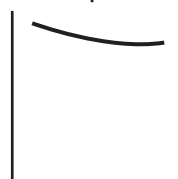
a Graph 1



b Graph 2



c Graph 3



d Graph 4



3 Alkanes

4 Increase by $CH_2 = 14$

5 a Increases

b Decreases

c Molecular mass of fuel decreases

d Molecular mass of fuel decreases

6 Less energy is stored in products.

7 a C + C b C + H 8 C-H bond

9 (CH_4) There are only C-H bonds, so more energy is stored.

10 a Methane b Methane

13 Enthalpy changes and chemical reactions

- 'change in'
- Change in energy content between reactants and products.
- + indicates reaction is endothermic;
- indicates reaction is endothermic
- Equations 1 and 6 5 $-286 + (+98) = -188$
- 2(Equation 1) + equation 4 + reverse of equation 5;
 $\Delta H = -891 \text{ kJ}$
- Much energy is released.
- Avoids dangerous reactions, saves chemicals
- Negative
- Carbon, hydrogen, methane

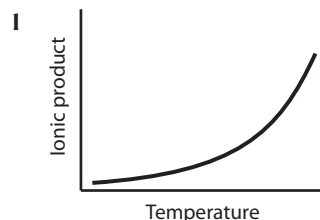
14 Chemical equilibrium

	No. of moles A_2	No. of moles B_2	No. of moles A_2B_4
Initially	4.0	3.0	0.0
At equilibrium	3.0	1.0	1.0

	$[A_2]$	$[B_2]$	$[A_2B_4]$
Initially	2.0	1.5	0.0
At equilibrium	1.5	0.5	0.5

- 50
- $K = \frac{[A_2B_4]}{[A_2][B_2]^2}$
- 1.33
- C
- a Increase b Decrease
- a Increase it b Decrease it
- 300°C
- Pushed to right, so K is increased.

15 Dissociation, ions and water



- Concentration of H^+ ions
- $[H^+] = (\times 10^{-7}): 0.33, 0.54, 0.82, 1.00, 1.21, 1.71, 2.37, 7.16$
- Water not a liquid outside this range.
- K_w increases with temperature, but not uniformly, etc.
- No
- $T \approx 60^\circ$
- $K_w \approx 2.0 \times 10^{-14}$
- Yes
- No, $[H^+] = [OH^-]$ always.
- Water dissociates only to a very small extent, particularly at room temperature.
- Poor conductor, few ions.
- Ability also increases.
- Increasing energy, more bonds broken to produce ions.

16 Dissociation of some binary compounds

- A chemical (liquid) that dissolves another substance.
- One which cannot dissolve any more solute at that temperature.
- (1) $[Cd^{2+}] \times [S^{2-}]$ (2) $[Fe^{2+}] \times [S^{2-}]$ (3) $[Co^{2+}] \times [S^{2-}]$
(4) $[Cu^{1+}] \times [Br^{1-}]$ (5) $[Cu^{1+}] \times [I^{1-}]$ (6) $[Cu^{2+}] \times [S^{2-}]$
(7) $[Ag^{1+}] \times [Cl^{1-}]$

- Chemical salts
- 4, 7, 5, 2, 3, 1, 6
- For each compound the metal ion valency equals the non-metal ion valency.
- No, the metal and non-metal combine in the ratio 2:3.
- They are all insoluble.
- Copper(II) sulfide (least), copper(I) bromide (most)
- Cu(I)Br best, Cu(II)S worst

17 Extraction of copper metal from ores

- a More thorough separation of gangue from ore.
b Bigger surface area for chemical reactions.
- Chalcocite, covellite
- Chrysocolla
- $Cu(OH)_2$ itself is a base.
- No
- Chrysocolla
- a 79.9 b 66.5
- Chalcocite
- 1.73
- a $FeO + SiO_2 \rightarrow FeSiO_3$
b $FeS + \frac{3}{2}O_2 + SiO_2 \rightarrow FeSiO_3 + SO_2$
- Gas
- Pollutant; forms acid rain.
- $Cu_2S + O_2 \rightarrow 2Cu + SO_2$
- Very strong
- Ionic
- H_2O, CO_2
- $4CuO + CH_4 \rightarrow 4Cu + CO_2 + 2H_2O$
- Oxidation
- $Cu(s) \rightarrow Cu^{2+}(aq) + 2e^-$
- Fall to bottom of cell container.
- Potential difference is applied to the cell.
- $Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$
- B

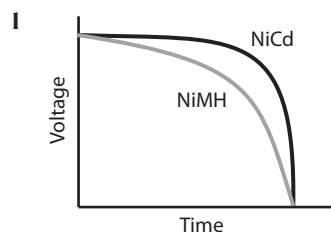
18 Infrared spectroscopy

- X contains A and C; Y contains B and C.
- a Chemical A b Chemical C
- Into the bonds. It causes increased vibration in the molecules which have more energy.
- Temperature increases.
- a No b C-C
- a Graph 1 b Graph 2
- No. A double bond is present in propene, but not in propane.
- Become more complex as bonds with different environments absorb different frequencies.
- a Both are EM vibrations and have same speed.
b They have different frequencies and wavelengths.

19 Electrolysis and Michael Faraday

- a 2 b 1.25×10^{19}
- 2.08×10^{-5} 3 2.496×10^{-2}
- $1.248 \times 10^{-2}Cu^{2+}, 2.496 \times 10^{-2}e^-, 1.248 \times 10^{-2}Cu(s)$
- 0.792 g 6 Same = 2.496×10^{-2} 7 $\approx 80 \text{ min}$
- a No, twice the number of moles of Ag since it is monovalent.
b Two moles Ag for every one mole Cu.
c No

20 Rechargeable batteries



	Ni-Cd	Ni-MH
Electrode undergoing oxidation	Cd	MH
Electrode undergoing reduction	Ni	Ni
Cathode	Ni	Ni
Anode	Cd	MH
Which electrode is +?	Ni	Ni
Which electrode is -?	Cd	MH

- 2 Alkali 3 Sodium/potassium hydroxide
- 4 a $\text{Cd} + 2\text{NiO}(\text{OH}) + 2\text{H}_2\text{O} \rightarrow \text{Cd}(\text{OH})_2 + 2\text{Ni}(\text{OH})_2$
b $\text{MH} + \text{NiO}(\text{OH}) \rightarrow \text{M} + \text{Ni}(\text{OH})_2$
- 5 $E^\circ = -0.71 \text{ V}$ for both 6 Lanthanides
- 7 Pollution aspects of other metals, i.e. lanthanides.
- 8 Each is a single unit, not a no. of units linked together which forms a 'battery'. Cell.
- 9 Refer to table for question 1.

21 Cells – users and producers of electricity

- 1 From Pt electrode; to Ni electrode

2 Table 1

Electrode	Pt/Ni	Charge	Half-reaction occurring	Oxidation/reduction	$E^\circ (\text{V})$
Anode	Pt	+	$2\text{Br}^- \rightarrow \text{Br}_2 + 2\text{e}^-$	Oxidation	-1.06
Cathode	Ni	-	$\text{Ni}^{2+} + 2\text{e}^- \rightarrow \text{Ni}$	Reduction	-0.25

- 3 $2\text{Br}^- + \text{Ni}^{2+} \rightarrow \text{Br}_2 + \text{Ni}$
- 4 $E^\circ = -1.31 \text{ V}$
- 5 Negative sign means not spontaneous.
- 6 Electrolytic
- 7 $V = +1.31$
- 8 Table 2

Electrode	Pt/Ni	Charge	Half-reaction occurring	Oxidation/reduction	$E^\circ (\text{V})$
Anode	Ni	-	$\text{Ni} \rightarrow \text{Ni}^{2+} + 2\text{e}^-$	Oxidation	+0.25
Cathode	Pt	+	$\text{Br}_2 + 2\text{e}^- \rightarrow 2\text{Br}^-$	Reduction	+1.06

- 9 Galvanic
- 10 1st, as emf puts energy into cell.
- 11 Decrease as Ni electrode goes into solution.
- 12 1st

22 Catalysts

- P1 Graph 1: uncatalysed; Graph 2: catalysed
- P2 Time and energy are both less in catalysed reaction.
- P3 No, only pathways change.
- P4 Yes
- 1 Transition metals

- 2 Transition metals have multiple valencies; Group 1 and 2 do not.
- 3 a $\text{SO}_2 + \frac{1}{2}\text{O}_2 \rightarrow \text{SO}_3$ b $\text{C}_2\text{H}_4 + \text{H}_2 \rightarrow \text{C}_2\text{H}_6$
c $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$ d $2\text{NO}_2 \rightarrow \text{N}_2 + 2\text{O}_2$
e $\text{CO} + \frac{1}{2}\text{O}_2 \rightarrow \text{CO}_2$
- 4 No 5 d and e
- 6 $2\text{NO} \rightarrow \text{N}_2 + \text{O}_2$ 7 Not enough time.
- 8 No, only higher activation energy.
- 9 Energy is absorbed, i.e. reaction is endothermic.
- 10 Hot gases rise. 11 $\text{C} + \text{O}_2 \rightarrow \text{CO}_2$
- 12 It lowers the activation energy, so reaction occurs at lower temperature.

23 Chemists and calculations

- 1 Transition metals 2 Contains no water.
- 3 a 6^+ b 3^+
- 4 Cr = 39.7%, Cu = 39.8%, B = 17.7%
- 5 325.0 g 6 140.9 g
- 7 0.455 M 8 $\text{H}_3\text{BO}_3 \rightarrow 3\text{H}^+ + \text{BO}_3^{3-}$
- 9 6.81 mol in 5.0 L 10 1.365 M
- 11 No. Greater in anhydrous form. No mass of water involved.
- 12 Colour due to chromium metal.

24 Cement – an important chemical in our society

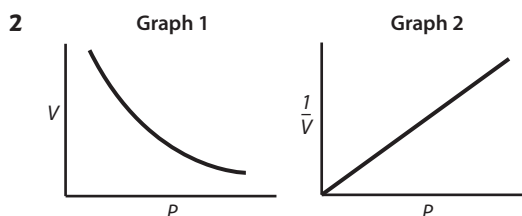
- 1 a CaO b MgO c Al_2O_3 d FeO e Fe_2O_3

No.	Formula	Formula written as oxides	Name of compound
1	CaSiO_3	$\text{CaO} \cdot \text{SiO}_2$	Calcium silicate
2	Al_2SiO_5	$\text{Al}_2\text{O}_3 \cdot \text{SiO}_2$	Aluminium silicate
3	MgAl_2O_4	$\text{MgO} \cdot \text{Al}_2\text{O}_3$	Magnesium aluminate
4	Fe_2SiO_5	$\text{Fe}_2\text{O}_3 \cdot \text{SiO}_2$	Iron(III) silicate
5	Fe_2SiO_4	$2\text{FeO} \cdot \text{SiO}_2$	Iron(II) silicate

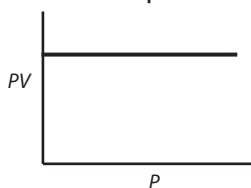
- 3 1:1 4 No
- 5 Endothermic 6 $34 \times 10^7 \text{ kJ}$
- 7 Extremely large
- 8 Landscape denudation, dust produced at all stages, large amounts of CO_2 formed, etc.

25 The effect of pressure on a sample of gas

No.	$\frac{1}{V}$	PV
1	0.408	248.19
2	0.613	247.76
3	0.813	249.20
4	1.020	248.23
5	1.220	249.20
6	1.450	247.23



Graph 3



- 3 It is the independent variable. 4 Hyperbola
5 Inverse relationship 6 Yes
7 a ≈ 230 kPa b ≈ 0.77 L
8 0.1 mole 9 a 4.0 g b He
10 The volume of a mass of gas (at constant temperature) is inversely proportional to the pressure acting on it.
11 $V \propto \frac{1}{P}$ or $P \propto \frac{1}{V}$ 12 Remains constant.
13 No.
14 Pressure causes gas to liquefy.
15 No. Applies only while it is a gas.
16 Care needed to avoid going beyond gaseous phase.
17 Gas has turned to a liquid which is not compressible.
18 Changed into heat energy.
19 Need to continually remove excess heat to keep temperature constant.
20 As the pressure on a mass of gas at constant temperature is increased, the volume decreases such that PV remains a constant.
21 It is a law because it always works with a gas.

26 The Solvay process

- 1 (Without phases)
a $2\text{NaCl} + \text{CaCO}_3 \rightarrow \text{Na}_2\text{CO}_3 + \text{CaCl}_2$
b A: $\text{NaCl} + \text{CO}_2 + \text{NH}_3 + \text{H}_2\text{O} \rightarrow \text{NaHCO}_3 + \text{NH}_4\text{Cl}$
B: $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$
C: $2\text{NH}_4\text{Cl} + \text{CaO} \rightarrow 2\text{NH}_3 + \text{CaCl}_2 + \text{H}_2\text{O}$
D: $2\text{NaHCO}_3 \rightarrow \text{Na}_2\text{CO}_3 + \text{H}_2\text{O} + \text{CO}_2$
2 B and D 3 Ammonia, water, carbon dioxide
4 a One b One c One
5 Yes 6 a Sea water b Limestone/marble

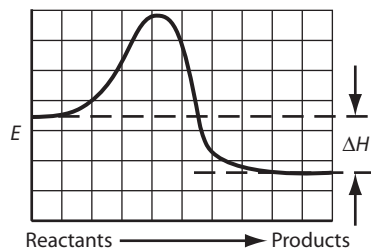
27 Desalination

- 1 B 2 a Less conc. b More conc.
3 Solution A: 2.5×10^{-2} ; Solution B: 5.0×10^{-3}
4 No 5 a $(50 + x)$ mL b $(50 - x)$ mL
6 a 0.3 M b 33.3 mL c A = 83.3 mL and B = 16.7 mL
7 0.6 M 8 1.2 M
9 Less salt because of freshwater input.
10 3500 g 11 a 233.3 g L^{-1} b 4.0 M
12 6.7 times 13 Sink to the bottom.
14 The conc. solutions could kill marine life.
15 Diluted with vast amounts of sea water.

28 The thermite reaction

- 1 $\text{Fe}_2\text{O}_3(\text{s}) + 2\text{Al}(\text{s}) \rightarrow 2\text{Fe}(\text{l}) + \text{Al}_2\text{O}_3(\text{l}); \Delta H = -853 \text{ kJ}$
2 $\text{Mg}(\text{s}) + \frac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{MgO}(\text{s}); \Delta H = -$

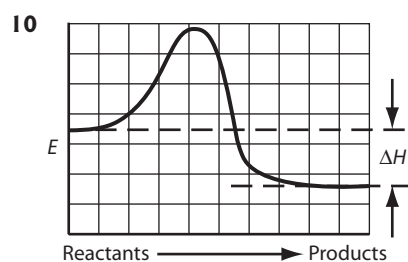
- 3 a Al b Fe_2O_3
4 a $2\text{Al} \rightarrow 2\text{Al}^{3+} + 6\text{e}^-$ b $2\text{Fe}^{3+} + 6\text{e}^- \rightarrow 2\text{Fe}$
5 Probably quite high.
6



- 7 They would melt, allowing mixing.
8 $\text{Fe}_2\text{O}_3 = 29.9 \text{ kg}$, $\text{Al} = 10.1 \text{ kg}$
9 $1.6 \times 10^5 \text{ kJ}$
10 a Leave the equation as is. b Reverse the equation.
c Add the two.
11 $\Delta H = -1670 \text{ kJ} + 820 \text{ kJ} = -850 \text{ kJ}$
12 The values are approximately the same.
13 The zinc or lead metals produced from the respective reactions of their oxides with aluminium may not form a strong bond with the iron of the rails or tram tracks and so not weld the tracks together successfully. In fact, they may not bond with the iron of the tracks at all.

29 Photosynthesis

- 1 $6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$
2 180 g 3 264 g 4 147.0 L
5 22.7 6 3.78 7 726.4 g
8 $6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \dots \Delta H = +2830$
9 Photosynthesis is an endothermic reaction.



- 10
11 a ΔH is negative.
b Respiration is an exothermic reaction.

30 Extraction of some metal ores

- 1 a Au, Ag, Pt, Hg, Cu b O, N, C, S, He, Ne, Ar, Kr, Xe, Rn
2 They do not react. 3 $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$
4 $3\text{MnO}_2 + 4\text{Al} \rightarrow 3\text{Mn} + 2\text{Al}_2\text{O}_3$
5 It is oxidised.
6 a $2\text{ZnS} + 3\text{O}_2 \rightarrow 2\text{ZnO} + 2\text{SO}_2$
b $2\text{PbS} + 3\text{O}_2 \rightarrow 2\text{PbO} + 2\text{SO}_2$
7 The two-step process uses less energy and is cheaper.
8 Na and K are very active. Their compounds are not easily altered. It is difficult to reduce Na^+ or K^+ ions.
9 a Cathode b Reduction
10 The water will be electrolysed in preference.

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